

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/371503006>

Satellite Communication: State-of-the-Art and Future Challenges

Article · June 2023

CITATIONS

6

READS

3,307

4 authors, including:



Muhammad Nauman Danish

Air University

10 PUBLICATIONS 16 CITATIONS

SEE PROFILE

Satellite Communication: State-of-the-Art and Future Challenges

Muhammad Nauman Danish, Syed Ahmed Pasha,

Abstract—Beyond line of sight communication is important as it provides, for example, tactical data sharing for military advantage. Satellite communication (SATCOM) is the only technology that provides this capability. Recent advances in micro-electronics technology have significantly enhanced the capability of SATCOM systems in terms of their ability to reprogram and reconfigure. Moreover, signal processing units can be simulated using software-defined radios. But a growing demand for high data rates and user anonymity, requiring implementation of higher-order modulation, multiplexing and efficient spread spectrum schemes, pose new challenges for SATCOM. These are compounded in the presence of reliability and low latency requirements for tactical communication as well as communication between various satellite technologies in the future such as constellations and 5G technology. In this survey, we review the state-of-the-art in SATCOM systems with respect to modulation, multiplexing, channel encoding and software-based implementations and discuss future challenges.

Index Terms—Satellite communication, high-throughput, software-defined radio, modulation, multiplexing, channel encoding

I. INTRODUCTION

SPACE technology has a huge impact on the economic and social aspects of both commercial as well as military domains. Driven by remarkable advances in the micro-electronics technology in recent years, building of smaller satellites by smaller teams has become a reality resulting in rapid development of multi-orbit satellite systems. According to an estimate¹, 2,666 operational satellites were circling the globe in April 2020 with > 70% in the low earth orbit. More than a third were communication satellites, > 15% were earth observing satellites and about 4% were navigation and positioning satellites. These have the potential to hugely impact a number of areas such as navigation, communication, weather prediction and disaster management while facilitating beyond line of sight communication in multiple orbits. Recent developments in satellite technology have opened the landscape for space communication with reprogrammable and reconfigurable capability offering low cost, fast, more frequent, and broader access to space. Satellite technology has vast military applications enabling network-centric battle force capability along with tactical data sharing between multi-generation aircraft.

M.N. Danish is with the Institute of Avionics and Aeronautics, Air University, Islamabad, Pakistan e-mail: (181904@students.au.edu.pk).

S.A. Pasha is with the Department of Electrical and Computer Engineering, Air University, Islamabad, Pakistan e-mail: (s.pasha@mail.au.edu.pk).

¹<https://www.weforum.org/agenda/2020/10/visualizing-earth-satellites-space-spacex/>

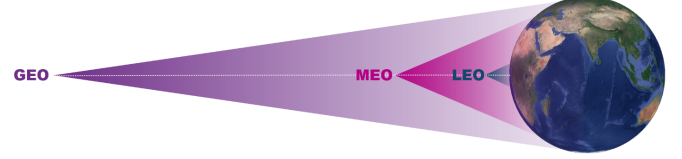


Fig. 1: Footprints of GSO and NGSO Satellites.

The rapid advancements in satellite technology have exploited multiple orbits for satellite communication (SATCOM) systems. To minimize aerodynamic drag, satellites are launched into space to orbit the earth at altitudes well above 300 km. In the geosynchronous orbit (GSO), a geosynchronous equatorial orbit (GEO) satellite is launched above the equator and orbits the earth at an altitude of 35,786 km with an orbital velocity matching that of the earth so that it appears at a fixed position in the sky. Thus, it can be used as a reliable communication relay as well as for providing broadcast services over a large geographical area. The non-geostationary-satellite orbit (NGSO) comprises multiple orbits at various altitudes, namely, low earth orbit (LEO) (< 2,000 km), medium earth orbit (MEO) (2,000-35,786 km), polar orbit (200-1,000 km) and highly elliptical orbit (HEO). Non-GEO satellites are available over a certain area on the earth for a brief duration only and therefore a group or *constellation* of satellites is needed for uninterrupted connectivity. Moreover, the ground segment requires a satellite tracking mechanism to communicate with non-GEO satellites if directional antennas are used. The coverage of GSO and NGSO satellites is illustrated in Fig. 1² and a comparison is summarized in Table I. Table II is a list of frequently occurring acronyms throughout the paper.

The basic architecture of a communication satellite involves a ground terminal, a space terminal, and equipment for communication protocols comprising a SATCOM modem, block up-converter (BUC), intermediate frequency (IF) cables, low-noise block (LNB), dish antennas and a centralized facility monitoring and controlling the communication network. Fig. 2 illustrates the basic architecture of a SATCOM system. The space terminal provides communication routing for the ground terminals. It has the capability to manage access control adaptively and in a multi-beam setup, implements the spot-beam management system [1]. The earth-based equipment (ground terminal) incorporates gateways with fiber-optic connectivity to provide end-user services. The (centralized) monitoring facility comprises multiple network control and management centers (HUB) that can provide real-time data services [2].

²<https://www.satellitetoday.com/contentcollection/ses-hub-geo-meo-and-leo>

TABLE I: Comparison of GSO and NGSO Satellites.

GSO ~36,000 km	NGSO: MEO ~8,000 km	NGSO: LEO ~1,000 km
High latency ~700 ms	Low latency ~150 ms	Very low latency ~50 ms
Very large footprint	Large footprint	Small footprint
Continental gateways (HTS for data)	Regional gateways (high throughput)	Many local gateways (low throughput)
Stationary antennas (3 satellites for global coverage)	1 hour tracking (6 satellites for global coverage)	10 minute tracking (~[100-1000] satellites for global coverage)
100 Mbps per terminal	Multiple Gbps per terminal	100 Mbps per terminal

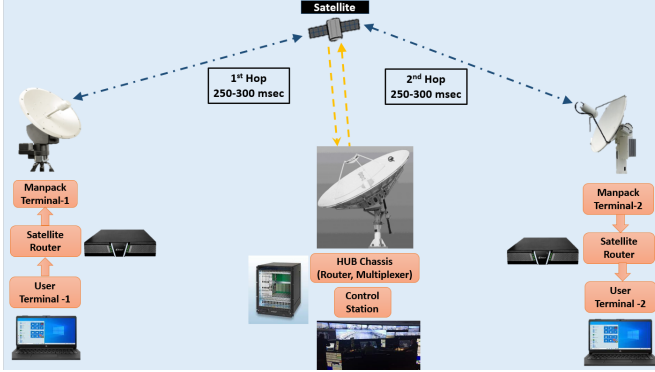


Fig. 2: Basic Architecture of a SATCOM System.

The HUB facility configures the network, allocates available resources, maintains network architecture and ensures user anonymity. The terminals in a SATCOM system comprise both fixed and mobile user ends that can access the SATCOM network directly or through terrestrial access points.

The SATCOM modem, a central component of the ground terminal, performs, for example, data conversions, modulation, channel encoding, multiplexing and communication protocols while converting base band to radio frequency (RF) band and vice versa. The processing units for SATCOM based on signal processing devices can be realized using a software-defined radio (SDR). Table III summarizes the characteristics of some commercially available SATCOM modems.

Technological developments in electronics have made it possible for satellite manufacturers and operators to consider digital payloads with on-board processing capability. Thus, limitations in ground processing for SATCOM systems are overcome by modern signal processing techniques. For example, delay in processing systems is mitigated by reducing processing and acquisition delays. Similarly, efficient resource utilization is enabled by scheduling capability and flexibility to integrate with new hardware [3]. These techniques have been used for on-board pre-distortion reduction and energy detection in [4], [5].

The weight and size of a satellite are important considerations for launching the satellite into space. Light-weight and small size satellites with advanced signal processing hardware is the modern paradigm leading to small satellite constellations and high throughput satellites (HTS) offering broadband internet-based applications with rapid media streaming for under-served areas. Moreover, seamless communication between 5G communication systems and small satellite constellations offering uninterrupted communication

services has the potential to integrate advanced wired and wireless systems and protocols for particular requirements and desired applications. This will usher in a new space era facilitating protocol development for mega-constellations, machine-to-machine communication (M2M), ultra-reliable low-latency communication (URLLC), and HTS systems.

The next generation of satellites demands an evolutionary design to address current challenges such as internet access to far-end users, geographical limitations of GEO, MEO and LEO satellites [6], advanced on-board processing capability, efficient tracking antennas to complement LEO satellite operations, robust inter-satellite links [7], real-time applications based on lower orbit satellite constellation operations to exploit smaller propagation delay [8], advanced routing and addressing protocols to ensure quality of service [1], software and hardware upgrades based on new communication and networking protocols to cope with new user requirements [9], [10], interoperability between multi-orbit satellites and operators [11], and seamless integration of existing satellite architecture and terrestrial network [12]. These challenges can potentially be overcome by operational capabilities of software-defined networking (SDN) [13], network virtualization (NV) [14], network function virtualization (NFV) [15] and SDR [16].

The existing body of work on advanced SATCOM systems [3], [17]–[33] is concerned with specific technologies for 5G and 6G networks and the role of satellites in these networks. In this survey paper, we aim to bridge the gap between developments in SDR-based ground stations, corresponding technologies and advancements in communication protocols with respect to multiplexing, modulation, and channel encoding in a structured way and highlight the main challenges, solutions and open problems. Moreover, we identify new research directions in advanced wireless communication technologies for future SATCOM systems. More specifically:

- We give a comprehensive treatment of multiple access schemes for SATCOM systems achieving interference mitigation and reduced complexity. We discuss hybrid schemes for satellite internet of things systems and advanced orthogonal frequency division multiple access, and non-orthogonal multiple access based GEO and LEO satellite network.
- We discuss the different fading phenomena encountered by geostationary (fixed) satellite systems and LEO satellites which can severely degrade the link margin and review various approaches to model channel propagation and attenuation characteristics in SATCOM systems. We review various studies on modulation schemes and give our recommendation for the GSO and NGSO SATCOM

TABLE II: List of Acronyms.

Acronym	Definition
ALOHA	additive links on-line Hawaii area
APSK	amplitude phase shift keying
BCH	Bose–Chaudhuri–Hocquenghem codes
BCJR	Bahl-Cocke-Jelinek-Raviv algorithm
BLER	block error rate
BP	belief propagation
CC	convolutional codes
CDMA	code division multiple access
CER	codeword error rate
CP	cyclic prefix
CRC	cyclic redundancy check
CSI	channel state information
DS-CDMA	direct sequence code division multiple access
DSL	digital subscriber line
DSSS	direct sequence spread spectrum
DVB-S	digital video broadcasting standard
eMBB	enhanced mobile broadband
EXIT	extrinsic information transfer chart
FDMA	frequency division multiple access
FEC	forward error correction
FH-CDMA	frequency hopping code division multiple access
FPGA	field programmable gate array
GEO	geostationary earth orbit
GFSK	Gaussian frequency-shift keying
GSO	geostationary orbit
GNSS	global navigation satellite system
HPA	high power amplifier
HTS	high throughput satellites
IoT	Internet-of-Things
ISI	inter symbol interference
ICI	inter carrier interference
LDPC	low density parity check
LEO	low earth orbit
MCSS	multi code spread spectrum
MEO	medium earth orbit
MF	matched filter
MF-TDMA	multi-frequency time division multiple access
mMTC	massive machine type communication
MODCOD	modulation and coding
MSK	minimum-shift keying
NOMA	non-orthogonal multiple access
NGSO	non-geostationary orbit
OFDMA	orthogonal frequency division multiple access
TIBWB-OFDM	time interleaved block windowed burst-OFDMA
GFDM	generalized frequency division multiplexing
F-OFDMA	filtered-OFDMA
OMA	orthogonal multiple access
OQPSK	offset quadrature phase-shift keying
PAPR	peak-to-average power ratio
PEG	progressive edge growth
PEXIT	protographic extrinsic information transfer
PC	polar codes
RF	radio frequency
SC-FDMA	single-carrier frequency division multiplexing access
SIC	successive interference cancellation
SCL	successive cancellation list
SDMA	space division multiple access
SDR	software-defined radio
SINR	signal-to-interference-plus-noise ratio
SPA	sum-product algorithm
SSMA	spread spectrum multiple access
SNR	signal-to-noise ratio
TB-CC	tail biting convolutional codes
TC	turbo codes
TDMA	time division multiple access
TPC	turbo product code
TWTA	travelling wave tube amplifier
URLLC	ultra-reliable low-latency communication

system based on performance, complexity and hardware compatibility at the RF frontend. Moreover, we discuss advantages of adaptive and variable modulation and channel coding for GSO and NGSO SATCOM systems.

- We present a detailed discussion on the state-of-the-art in SATCOM channel encoding and the related challenges with a particular focus on codes for the short block-length regime. Moreover, we draw performance comparison for variant of LDPC codes, Turbo codes, and Polar codes.
- We review developments in SDR for multiple orbit SATCOM systems with emphasis on processing and transmission capabilities and limitations of SDR-based approaches. We discuss the development platforms. The SDR based architecture for adaptive satellite transceivers for GSO and NGSO systems and their role in future wireless communication systems are also discussed.

The remainder of the paper is structured as follows: Section II is concerned with channel access techniques for SATCOM systems. In section III, we discuss modulation schemes and channel modelling. Section IV focusses on channel encoding schemes for wireless and satellite communication systems and section V discusses SDR-based approaches. A discussion on next generation wireless communication systems is presented in section VI. Section VII concludes the paper. Fig. 3 is a graphical representation of the structure of the main parts of this survey paper.

II. MULTIPLE ACCESS CHANNEL MODELS FOR SATCOM

One of the primary functions performed by a SATCOM modem is multiplexing. For high throughput, bandwidth sharing approaches based on frequency division multiple access (FDMA), time division multiple access (TDMA), code division multiple access (CDMA), spatial division multiple access (SDMA), orthogonal multiple access (OMA) and non-orthogonal multiple access (NOMA) are used. FDMA is the simplest to implement but requires allocation of distinct frequency bands to each subscriber. TDMA is effective but requires framing and synchronization which is problematic in environments with significant propagation delay. CDMA shares the available band, based on pseudorandom patterns for multi-user environments, by assigning unique random patterns to each subscriber. This allocation enables the use of multiple codes on the same frequency channel for different users. At the receiver, segregation of the codes can be achieved using a correlator and filter to despread the signal for the intended user. Typically, CDMA is coupled with a spread spectrum technique, e.g., direct sequence (DS)-CDMA and frequency hopping (FH)-CDMA [34]–[40]. DS spreading entails multiplication of information bits with a pseudorandom pattern. In FH spreading, the bandwidth is distributed into multiple sub-frequency bands. Then, information is transmitted based on a certain hopping pattern. The hopping pattern results in different channels and interfering signals for the intended transmission in each hop, avoiding the unintended signal. FH-CDMA consumes less power and requires low budget equipment while DS-CDMA has shown greater reliability and efficiency. Moreover, it has outperformed modern multiplexing

TABLE III: Specifications of Commercial SATCOM Modems.

Modem	Topology	Modulation	FEC	DSSS	Interfaces
X5-Idirect	DVB-S2(DS), iNFINITI-TDM(DS), A-TDMA(US), SCPC Return(US)	QPSK, 8PSK, 16APSK(DS), BPSK, QPSK, 8PSK(US) Max Symbol Rate (in Msps) 45 or 15/7/5 or 15	LDPC, TPC 2D-16 CC Code	up to 7.5 Mcps SF: 1,2,4,8	Tx= 0.95–1.7 GHz Rx= 0.85–2.15 GHz AES-256 encryption 10 MHz reference controllable on Tx/Rx IF port
X7-Idirect	DVB-S2(DS) TDMA(US)	QPSK, 8PSK, 16APSK(DS), BPSK, QPSK, 8PSK(US) Max Symbol Rate (in Msps) 1–45 & 0.128–7.5	LDPC (1/4-8/9) 2D-16 state with (1/2–6/7)	upto 7.5 Mcps SF: 1,2,4,8	Tx= 0.95–1.7 GHz Rx= 0.95–2.15 GHz
Zenith Modem	Not Given	BPSK, QPSK, OQPSK, 8PSK, 18QAM, 16QAM	Extended LDPC RS, TPC	N.A.	N.A.
9800 AE	DVB-S2 A-TDMA & ACM	QPSK, 8PSK, 16APSK	LDPC(1/4–8/9)	up to 15 Mcps SF: 1,2,4,8,16	Tx= 0.95–2.05 GHz Rx= 0.95–2.15 GHz

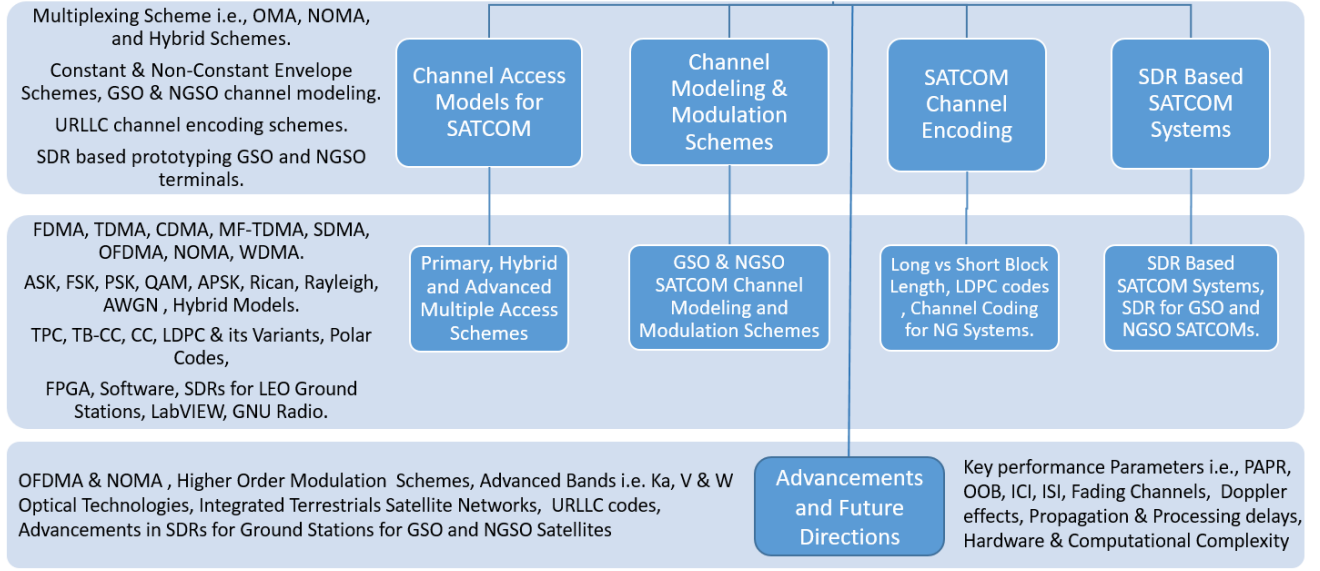


Fig. 3: Organization of the Paper.

schemes such as orthogonal frequency division multiplexing (OFDM) and multi-code spread spectrum (MCSS) in terms of anti-jamming capability and reliability [41]. Fig. 4 illustrates various multiplexing schemes. A detailed discussion follows below.

A. Primary Multiple Access Schemes

For modems using FDMA, TDMA, and CDMA for SATCOM, interference problems are mitigated if separate frequency bands for the uplink and downlink are used [42]. As already mentioned, FDMA is limited by the requirement of separate frequency bands, TDMA requires strict synchronization protocols, i.e., frame synchronization, carrier recovery, symbol recovery, loop-back block, orbital effects catering block and other synchronizing blocks, while CDMA is limited by bandwidth expansion factors and low data rates.

Spread spectrum multiple access (SSMA) with binary phase shift keying (BPSK) and orthogonal convolutional codes (CC) has shown that signal spreading using forward error correction (FEC) coding with code rates ≤ 1 fully exploits the processing gain and achieves the ultimate Shannon capacity for multiple access channel models [43]. However, this performance is only achieved for very low rate FEC codes having large bandwidth

expansion of multiples of 2^K with pseudorandom sequences where K is the sequence constraint length. For dedicated channels and a reasonable signal-to-noise ratio (SNR), SSMA has a lower bandwidth efficiency than FDMA but unlike FDMA, is not limited by the requirement of dedicated channels.

Due to the random access nature of CDMA coupled with reasonable degradation and appropriate ranging methods, BPSK-DSSS has been proposed for ground-satellite mobile stations with L-band frequency and 3 MHz channel bandwidth [44]. Fades of 3-7 dB due to the presence of trees and deep fades of 15 dB due to tall buildings and tunnels have been reported.

DSSS vs. FDMA: DSSS has exhibited remarkable performance to combat fading due to multipath propagation effects in cellular communication systems [36]. A study of the DSSS-CDMA system with frequency selective fading channels, more specifically, 4-state Rician and 4-state shadowed distributions for intermediate circular orbit satellites has shown that quadrature phase shift keying (QPSK) with the raised cosine filter and an adaptive receiver based on the least mean square (LMS) algorithm is insensitive to timing and phase synchronization [34]. Soft decision decoding leads to an improvement in the performance under fading conditions. Moreover, an adaptive

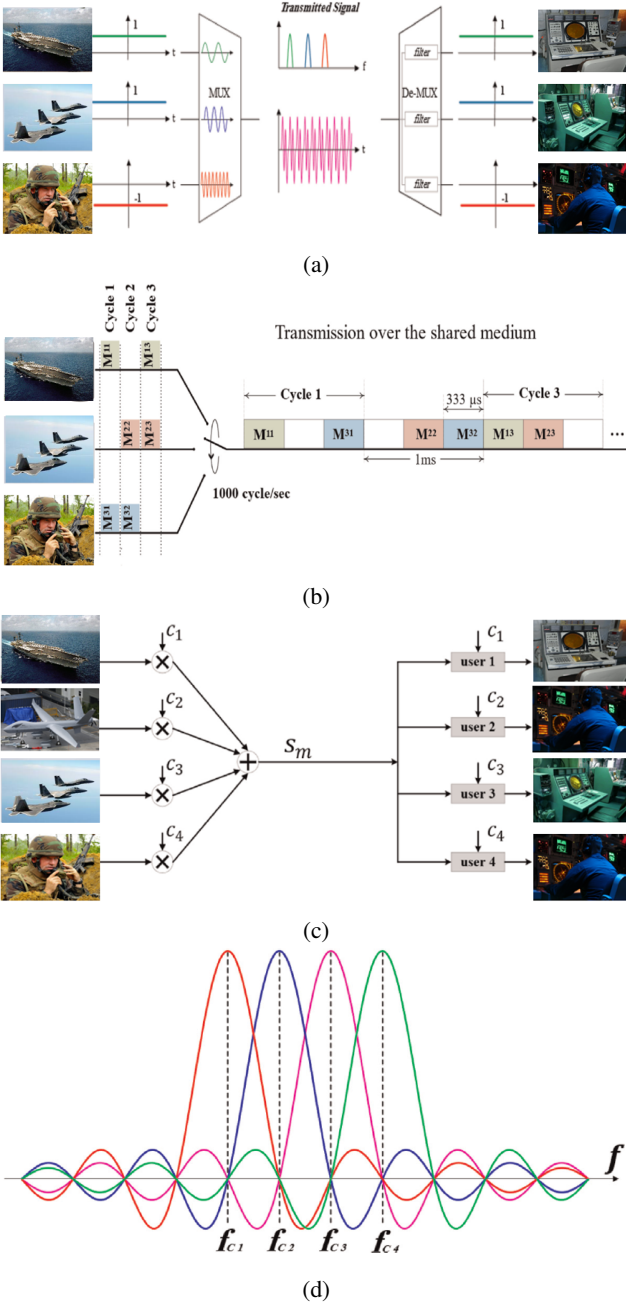


Fig. 4: Multiplexing Schemes: (a) FDMA, (b) TDMA, (c) CDMA, (d) OFDMA. [20]

linear receiver increases system utilization up to 70-80% at the cost of a moderate increase in complexity as compared to the conventional matched filter (MF) with channel capacity limited to 10-20%. For maximum efficiency, low rate CC may be avoided by employing an adaptive receiver.

For very long and power limited high frequency (3-30 MHz) links, the performance of DSSS and OFDM has been studied in terms of the error and differential error vector magnitude [45], [46]. Spectral efficiency of DSSS is inversely proportional to the length of the pseudonoise (PN) sequence. For OFDM, a soft limiter is applied to check the peak-to-average power ratio (PAPR) to avoid operating in the nonlinear

region of the amplifier. Experimental studies have shown that the longest symbol for DSSS does not guarantee the lowest bit error rate (BER) because the long symbol time exceeds the channel coherence time resulting in high phase rotation. Small symbols lead to large variations in the channel impulse response. Thus, an intermediate symbol length gives the best results. For OFDM, channel encoding and interleaving are required to reduce the BER. The data rate for OFDM is 10-15 $\times$ higher compared to DSSS, resulting in 20-40 $\times$ higher spectral efficiency. But in poor SNR conditions with the data rate below the actual channel capacity, DSSS can transmit data without channel encoding and interleaving. Moreover, DSSS favours longer period transmissions below the noise floor, reducing interference from other users.

DSSS with multi code spread spectrum (MCSS) for wireless video transmission has been studied in [41]. The system was realized using an SDR platform and shown to outperform OFDM in terms of anti-jamming capability. However, this is accomplished at the cost of bandwidth expansion for data transmission. With a lower PAPR, DSSS yields higher power efficiency compared to OFDM with MCSS. The findings in [41] can be summarized as:

- (i) DSSS outperforms OFDM with MCSS in terms of anti-jamming capability but OFDM has a higher spectral efficiency.
- (ii) MCSS performs moderately from both aspects, i.e., shows lower spectral efficiency than OFDM and lower bandwidth expansion than DSSS. MCSS shows better anti-jamming performance than OFDM but not DSSS.

To ensure user anonymity, a modern concept of protected tactical waveform (PTW) for military wideband communication has been analyzed in [47]. This approach uses FHSS-based frequency diversity, transmission security driven time randomization, and interleaver-based time diversity and concatenated Bose–Chaudhuri–Hocquenghem (BCH) and low-density parity-check (LDPC) codes. MPSK modulation schemes for different code rates with code length $n > 1000$ have shown that QPSK with $\frac{1}{4}$ code rate can achieve a block error rate (BLER) of 10^{-4} within 0.1 dB. Higher modulation and coding (MODCOD), e.g., 8PSK with $\frac{8}{9}$ code rate requires 2.5 dB to attain the same bound.

The digital video broadcasting-return channel via satellite (DVB-RCS2) standard for SATCOM systems optionally supports random access protocols for return link communication. ALOHA, a popular random access protocol, supports transmission from nodes ignoring channel conditions but requires an acknowledgment (ACK) from the target terminal. If ACK is not received, the same packet is retransmitted after a time-out period. ALOHA is supported by recent internet of things (IoT) technologies which include LoRa and SigFox [21].

Properties of the FHSS, DSSS, and OFDM spread spectrum schemes are summarized in Table IV.

B. Hybrid Multiple Access Schemes

The choice between a fixed and a random assignment mechanism in a network depends on the type of traffic.

TABLE IV: Properties of the FHSS, DSSS and OFDM Spread Spectrum Schemes.

Spread Spectrum	Features
FHSS	• Multiple frequency channels • Pseudorandom sequence • Rapid switch (hopping) • Interference for short interval only
DSSS	• Higher bandwidth • Spreading using pseudorandom sequence • Orthogonal codes except intended receiver • Most widely used type of CDMA • Better anti-jamming performance with larger bandwidth compared to OFDM and MCSS • Better performance in noise compared to FHSS
OFDM	• Higher data rate • Inter-carrier interference limitation • PAPR limitation to avoid transmitter saturation • Better spectral efficiency compared to DSSS & MCSS

For example, for bursty traffic, a fixed assignment results in inefficient use of spectrum resources. The demand assignment multiple access (DAMA) underperforms in sporadic IoT traffic having very short packet length and low duty-cycles [48], [49]. In such environments, random access protocols are preferred which transmit data packets irrespective of the nodal and channel conditions but can lead to packet collisions. Fixed assignment protocols allocate fixed resources to each user in the network with respect to time and frequency based on the user requirements. Thus, even in simultaneous transmission from multiple users data packet collisions are avoided. Examples include OFDMA and single-carrier (SC)-FDMA. However, strict synchronization in time and frequency is required to avoid co-channel interference. In narrowband (NB)-IoT, OFDM and SC-FDMA are used for the downlink and uplink communication respectively [50]. Similarly, a chirp spread spectrum signal is employed in LoRa [51] and an ultra-narrowband signal with differential BPSK in SigFox.

The traffic in a satellite system is controlled by the network control center (HUB) which monitors and distributes resources to meet the demands from the return channel satellite terminals (RCSTs). The use of hybrid schemes, e.g., multi-frequency (MF)-TDMA allows RCSTs to control digitally modulated carriers by applying frequency hopping and transmit short bursts of data during the assigned time slots for different users. The alternative is to use a continuous carrier at the cost of increased complexity of the RCSTs. CC is more dynamic in terms of adapting to varying transmission requirements as compared to MF-TDMA. Multi-carrier modulation schemes such as OFDM and SC-FDMA or MF-TDMA are implemented in modern day satellite routers. In broadband satellite systems where the digital video broadcast standard is used, MF-TDMA has shown suboptimal performance [48].

For NB-IoT, the time-frequency ALOHA (TFA) has been proposed in [52]. But for IoT-based satellite applications, the significant propagation delay (250-270 ms) makes ALOHA unattractive despite performing well at very modest traffic requirements [53]. For satellite-IoT systems, advanced random access schemes have been discussed in [54]. The enhanced spread spectrum ALOHA (E-SSA) [55], asynchronous contention resolution diversity ALOHA (ACRDA) [56], and con-

tention resolution diversity ALOHA (CRDSA) [57] yield better spectral and energy efficiency due to the iterative successive interference cancellation (SIC) capability which improves the probability of detection of the received packets.

C. Advanced Multiple Access Schemes

Orthogonal Frequency Division Multiple Access: OFDM is the preferred scheme in multiple communication standards, e.g., DVB, DSL, WLAN, WMAN and 4G wireless networks. It consists of N adjacent orthogonal sub-carriers. The spacing between the sub-carriers depends on the Doppler effect, phase noise, radio-channel frequency selectivity and rate of channel variations. The sub-carrier spacing results in efficient demodulation at the receiver side due to reduced inter-carrier interference (ICI). The inverse fast Fourier transform (FFT)-based algorithm reduces implementation complexity at the transmitter side, and assists frequency domain equalization at the receiver. OFDMA is used in 5G New Radio communication systems, i.e., mobile broadband systems with the addition of cyclic prefix (CP) [58], similar to 4G LTE systems [24]. The implementation parameters for OFDMA are sub-carrier spacing, number of sub-carriers, and CP length [59].

SATCOM suffers from multipath fading. A potential approach to mitigate this phenomenon is the space frequency block code (SFBC)-OFDM. Somasekhar et al. [60] has discussed an algorithm to reduce ICI based on time selectivity and inter-symbol interference (ISI) induced by frequency selectivity and furthermore discussed the energy efficient resource utilization and numerical efficacy of the system. In [61], an SDR-based so-called network coded cooperation (NCC)-OFDMA has been demonstrated using three nodes capable of transmitting and receiving data. The combination of NCC and OFDMA has been shown to improve robustness and reliability of the wireless communication system against channel fading impediments. The performance has been evaluated using BER and error vector magnitude.

Filtered OFDMA (F-OFDMA), in addition to CP, involves spectrum shaping in each sub-band to reduce out-of-band (OOB) transmissions and uses guard intervals to suppress inter-sub-band interferences. However, it suffers from loss in time domain orthogonality, inner-sub-band interferences, and increased complexity [62]–[64]. Generalized frequency division multiplexing (GFDM), in addition to N sub-carriers, assigns different time slots for N_s sub-symbols, resulting in additional degree of freedom with $N_s N$ modulated constellation symbols [65], [66]. GFDM has the advantage that it requires only one CP for the entire transmission block resulting in higher spectral efficiency, and has lower OOB emissions than OFDM. However, it is prone to ISI and ICI, high PAPR and high implementation complexity [18], [67]. Multiple equalization and interference cancellation techniques have been proposed, e.g., matched filtering, zero forcing, double-sided SIC, SIC [66]–[72] to reduce the interferences between sub-carriers and time slots. Multiple architectures have been proposed to reduce the complexity for future wireless communication systems [67], [73]. Time-interleaved block-windowed burst OFDM (TIBWB-OFDM) implements

power spectral density based spectral confinement and windowing techniques to lower OOB emissions resulting in higher spectral efficiency than cyclic prefix-OFDMA (CP-OFDMA). Moreover, instead of CP, it employs concatenation of symbols with addition of single prefix of zeros. Thus, requiring lower power for transmission resulting in increased power efficiency [74]–[76]. However, the improvement in spectral efficiency is limited by the size of the OFDM windowed-based blocks. As the average transmitted signal power is reduced due to windowing operations, the overall PAPR grows [77].

In [78], the intentional interference mitigation capability for wireless systems has been studied and a secure pre-coded (SP)-OFDM system has been proposed for environments where a jammer can cause communication failure. The synchronization performance in additive white Gaussian noise (AWGN) and multipath fading channels has shown that SP-OFDM is superior to traditional OFDM and achieves positive capacity compared to zero under disguised jamming. For traditional OFDM systems in AWGN and Rician channels, the BER cannot be reduced by decreasing the code rate if proper anti-jamming procedure is not adopted. But in SP-OFDM, the BER can be significantly reduced with a decrease in the code rate. Both traditional OFDM and SP-OFDM show high spectral efficiency.

Besides the choice between fixed assignment and random access protocols, efficient use of the available spectrum can also be achieved using the dynamic spectrum sharing (DSS) technique which enables spectral coexistence of multiple wireless communication systems in the available radio spectrum. The DSS approach implements coordinated spectrum sharing, e.g., TV whitespace database or coordinated spectrum sensing. A study on DSS models [79], has discussed three types: the common model, the shared-use model and the exclusive use model. In the common model, all users have equal rights in terms of spectrum access. In the exclusive use model, secondary users have dedicated privilege to use the spectrum and in the shared-use model, users utilize the vacant and under-utilized spectrum in interference avoidance or opportunistic manner [80]. In heterogeneous networks, spectrum sharing is based on advanced mechanisms, e.g., licensed assisted access (LAA), channel bonding (CB), licensed shared access (LSA), carrier aggregation (CA) and spectrum access system (SAS) [17], [81].

Non-Orthogonal Multiple Access: Non-orthogonal multiple access (NOMA) is a promising technique for channel access in 5G systems [82]. Unlike OMA, it allows multiple users to share the same time-frequency resources. The co-channel interference resulting from reuse of the spectrum by multiple terminals can be mitigated using SIC and a multi-user detection system at the receiver station. Thus, increasing spectrum utilization with respect to the available number of users in 5G terrestrial scenarios [82]–[85]. The performance of NOMA in the 5G terrestrial-satellite network with respect to resource optimization, beamforming and power utilization has been studied in [86]. In [87], the power coefficients and beamforming weight vectors have been obtained to achieve a desired performance.

In [88], NOMA has been proposed for the Integrated

Satellite-Terrestrial Communication Networks (ISTCN) in 6G networks to overcome congestion and interference issues due to the use of common bands. 6G networks aim to provide a high speed and persistent network using efficient spectrum management and dynamic spectrum sharing with the help of NOMA and cognitive radios. The proposed network and network access model comprise HEO and LEO satellites with terrestrial cellular stems based on a hybrid network access model. For smooth spectrum sharing between the space and ground elements, user terminals are grouped at the satellite gateway based on the received CSI for uplink communication. For downlink, SIC is used for the receiver terminals. Opportunistic spectrum access is promoted for satellite users to suppress possible interference, i.e., based on available information, if a given channel is not in use by the terrestrial system, a satellite user may occupy it; otherwise, the user needs to switch to an idle channel. This is achieved using an interference cognition technique that can identify the interferences and provide anti-interference decision. Furthermore, power control using cognitive radios and satellite beamforming can reduce potential interference and congestion of spectrum in ISTCN.

Proactive content caching in hybrid satellite aerial networks (HSATN) can reduce transmission delay, improve spectral utility, and support wider connectivity for next generation wireless networks [89]. The system model for HSATN comprises the terrestrial systems, aerial platforms and satellite spacecrafts. The request for content from multiple users can be accommodated by the cache enabled aerial relays. The content caching model is based on content popularity skewness while the aerial platforms maintain finite content catalogue. However, if the requested file is not cached, the satellite system will directly accommodate the request using NOMA. During the content delivery phase, NOMA allows satellites to push the required content to the aerial relay.

For uplink communication, NOMA can reduce the frequency interference and coordination problem in multi-layer satellite networks (MLSN) [90]. MLSN can exploit the high throughput capacity of GEO satellites and the short delay in LEO constellations to meet 5G network requirements. A two layer system model for uplink communication has been proposed employing LEO and GEO satellites, with better channel condition than for LEO satellites alone. The N NOMA groups based on the user pairing scheme comprise a single GEO and multiple LEO users. The pairing is based on a *max-min* strategy to maximize the minimum channel difference between user pairs. The path difference between the GEO and LEO satellites creates an effective power attenuation. Thus, the GEO satellite can decode the GEO users' information without SIC. However, for LEO satellites SIC is used to decode the intended information.

Multi-beam capability offers users to reuse the available spectrum in a much more efficient way by exploiting the spatial characteristics of transmission and/or reception of antenna arrays. For example, the active electronically scanned array antenna delivers a cutting edge high throughput SATCOM system based on SDMA. But it requires advanced digital signal processing techniques to mitigate multi-user interference to achieve the desired performance. Such SATCOM systems

lead to multi-user, multiple-input and multiple-output (MIMO) systems (also commonly known as multi-beam joint processing systems) [3]. The linear beamforming (also known as *precoding*) method effectively avoids multi-user interference [91]–[94]. Extension of beamforming techniques for managing interference at SATCOM gateway systems is a recent development in broadband multi-beam SATCOM systems [95]. Beamforming for designing multiple data streams within the same bandwidth, takes advantage of channel state information (CSI) to mitigate multi-user interference for user terminals resulting in high signal-to-interference-plus-noise ratio (SINR) for specific users. An alternative to precoding multiple data streams is symbol level precoding based on CSI and data information of transmitted symbols [96]. The idea is to control the interference effects to produce constructive results at each user terminal resulting in reduced power consumption for a given SINR as compared to traditional techniques.

To meet the internet broadband services demand for non-uniform and distributed users, i.e., aeronautical, fixed satellite services users and maritime users, the adaptive multi-beam high throughput satellite system has been proposed [33] based on the four colour frequency reuse scheme [97]. The neighbouring cells have different carrier frequency or polarization and the space segment can alter the beam size and position using a programmable payload. The beam patterns are elliptical which add more freedom for optimization and are approximated by a two-dimensional Gaussian function. The broadband users are clustered for geographical grouping based on the weighted k -means clustering algorithm. For coverage and beam borders identification, the Voronoi tessellation is used with irregular polygons approximated by ellipses. The proposed method distributes the traffic across all beams equally resulting in better frequency and payload allocation, and radio resource management. Angular domain design parameters have been used for satellite beam pattern and footprint parameters.

To summarize, FDMA suffers from interference and requires separate bands for each user which reduces spectral efficiency. It does not scale with the number users due to the requirement of guard bands and adjacent channel interference. While TDMA is effective, it requires frame and orbital effects synchronization, carrier and symbol clock recovery, and loop-back synchronization which make it difficult to implement in the presence of significant propagation delay and diversity of user terminal. SDMA offers better spectral efficiency and synchronization complexity but requires the reuse distance in the network architecture to be preserved. CDMA can accommodate multiple users within the same bandwidth without the need for synchronization, is well-integrated with cellular technologies, and does not suffer from interference due to separate code word allocation. But these benefits come at the cost of higher complexity, and the need for guard band and guard time. Moreover, the overall quality of services can deteriorate if the number of users is increased. The performance of the variants of CDMA, namely DSSS and FHSS can vary. In the multipath and intentional interference environment, DSSS outperforms FHSS in the presence of noise but is limited by high acquisition time, wideband characteristics, and

the requirement of sequence generation at high rates. DSSS with an adaptive receiver instead of the conventional matched filtering enhances capacity of the system. A low rate spread spectrum technique for poor channel forecast can increase the integrity of the received data, whereas the multi-carrier OFDM technique for sporadic burst of data at higher rates when the channel forecast is good increases the transmission data rate. MCSS performance shows a trade-off between reliability and data rate. In comparison to OFDM, DSSS and MCSS effectively resist narrowband interference at the cost of bandwidth expansion. The advantages and limitations of some multiple access schemes for SATCOM systems achieving interference mitigation and reduced complexity are summarized in Table V. A suitable methodology can be selected based on the intended application of the SATCOM system.

For SATCOM systems, utilization of the available spectrum depends on the type of traffic used for data transmission. Both fixed and random access protocols can be used for services each with its advantages and limitations. For sporadic and modest transmissions random access protocols, e.g., ALOHA, E-SSA, ACRDA and CRDSA can provide better efficiency at the expense of packet collisions. Fixed assignment protocols, e.g., MF-TDMA and SC-FDMA are suitable for continuous and large packet transmission, avoiding data packet collisions at the expense of spectrum efficiency. Dynamic spectrum sharing capability is recommended to use the under-utilized and vacant spectrum. Similarly, multi-beam capability can enhance capacity of the SATCOM system in terms of high throughput and spectrum sharing used in, e.g., OMA and NOMA. NOMA has been studied for satellite-terrestrial integrated networks. However, both NOMA and OMA need to mitigate co-channel interference for successful sharing of the available spectrum. The use of NOMA along with SDR can create the opportunity for dynamic spectrum sharing that can further enhance utilization of the available spectrum.

III. SATCOM CHANNEL MODELLING AND MODULATION SCHEMES

Satellites in the geostationary orbit are ideal for SATCOM. A single geostationary satellite (GeoSAT) has greater coverage than medium and low earth satellites. Multi-beam capability enables the satellite system to reuse the frequency which results in higher throughput as compared to terrestrial networks. But the delay in GeoSATs precludes telephonic services for users. Constellations in the LEO orbit with cutting edge communication technology and modest launching overhead make LEO satellites a viable option for low latency and consistent coverage for SATCOM services. This is evident from the growing number of companies such as Amazon, TeleSAT, SpaceX, and OneWeb taking this approach [30]. SpaceX, had contributed 242 satellites to the Starlink constellations by January 2020. Similarly, O3B has launched a constellation comprising 20 MEO satellites above the equator with steerable antennas.

Combination of satellites in different orbits results in the formation of hybrid systems with more advanced communication capabilities and greater coverage. For example, the

TABLE V: Properties of Some Multiple Access Scheme.

Multiple Access	Advantages	Limitations
FDMA	- Simplest implementation - Multiple Variants i.e., SC-FDMA, OFDMA - SC-FDMA reduces PAPR - OFDMA could increase the spectral efficiency	- Separate frequency band - Crosstalk - Interference - Transmitter interruptions - SC-FDMA not suitable for higher data rates and large number of users - OFDMA suffers from ISI, ICI, PAPR and CP
TDMA	- Sharing without interference - Time slot based transmission	- Exact synchronization required - Inefficient capacity utilization - Wastage of Spectrum during user inactivity - complex hardware implementation design
SDMA	Frequency reuse	Reuse distance
CDMA	- Single channel for multiple users - Unique codes - Correlator - Matched filters	- Perfectly uncorrelated signal - Bandwidth expansion
OMA	Uncorrelated reuse signals	- Orthogonality - Co-channel interference
NOMA	- Precoding techniques reuse signals - Enhanced spectral efficiency - Support higher throughput	- Co-channel interference - Requires successive interference cancellation - Requisite propagation models based on SATCOM channel characteristics - Channel sharing based on channel gains - Design constraints

combination of MEO and GEO satellites and backhauling of LEO satellites using MEO and GEO satellites offer variable link and adaptive coding and modulation techniques. These hybrid systems make it possible to use multiple modulation schemes and adaptive MODCOD schemes to take advantage of varying SNR due to changing channel conditions to achieve higher throughput.

A. GSO and NGSO SATCOM Channel Modelling

For SATCOM, the communication channel is crucial due to the considerable distance between the transmitting and receiving terminals. The characteristics of the medium are not uniform between the terminals and depend on the frequency range used for the communication system. For fixed satellite systems, the channel is typically modelled for C-band (4-6 GHz). But for modern and future satellite systems, higher bands such as the Ku-band and Ka-band are preferred. The propagation model and attenuation characteristics for these three bands are not the same resulting in different channel models with different data rate, range, and reliability of transmission. Consequently, to study the channel characteristics, multiple models including the AWGN, hybrid fading, log-normal fading, Rician and Rayleigh fading models need to be considered [34], [43], [44], [98]–[102].

The fixed satellite channel can be modelled as AWGN [98] with first-order statistics modelling various atmospheric fading phenomena such as tropospheric scintillation, gaseous absorption and clouds, attenuation due to precipitation and signal depolarization. For SATCOM systems, the link margin is critical as it ensures the availability of a link in poor weather conditions. The effect of rain on the SATCOM link can be modelled using dynamic channel effects that characterize the temporal properties of the AWGN channel. Second-order statistics describe fade slope and duration. The relative distance between the ground terminals and the satellite compared to the onboard antenna distances leads to correlation in the fading and thus adversely affects MIMO systems [30].

For fixed SATCOM systems, the distance between the ground terminals needs to be significant to ensure de-correlation of fading in the rain [98], [103].

Modelling the satellite channel as a nonlinear dual link has shown that for synchronous transport module-level-1, a signal with 8PSK modulation scheme in 6/4 GHz frequency band can provide a link margin of 3 dB or more [104]. BCH codes (255,239) deliver an increased data rate of 160.6 Mbps and reduce the carrier-to-noise ratio required for a desirable BER. But at higher bands (14.5/12.5 GHz), due to the high fade margins and greater rain susceptibility, the above link margin is not sufficient. Transmission fading due to weather impairments and multipath fading can be modelled as a Gaussian or a log-normal distribution [105].

LEO satellites encounter a mobile channel that has different Rician and Rayleigh channel characteristics than a fixed channel. These channel effects are modelled using narrowband and broadband channel models involving multi-state Markov models. The narrowband model determines three characteristics of the channel, i.e., good state, deep and intermediate shadow state [106]. An alternative 2-state model assumes a Loo distribution for the received signal and channel state characteristics based on user terminal speed [107]. Similarly, for the broadband channel, shadowing is modelled using a 2-state semi-Markov model and multipath propagation is modelled using ITU multi-tap models [108]. In general, these models assume that the channel is quasi-stationary over short periods and can be treated as a stationary stochastic process.

The congestion in existing bands makes Ka-band a viable option for SATCOM due to its anti-jamming capability, capacity, bandwidth, and smaller size requirement for the terminals [100]. Multipath and shadow fading are the main sources of signal fading in poor weather conditions. The SATCOM signal can be modelled using the Rician distribution (line-of-sight component with multipath signals and amplitude fading characteristics). There exist a number of channel models for the Ka-band SATCOM system, each with its limitations. A hybrid

model based on normalization of the multipath and shadowing effect is preferred. It comprises normalized Rayleigh and log-normal distributions with multiplicative Gaussian distributions for surround and weather fading. The BER performance of MPSK (BPSK, QPSK and 8PSK) has been studied for the hybrid model. Simulation results have shown that BPSK is the most error tolerant while 8PSK has the highest transmission efficiency.

In a laser communication system for SATCOM, scattering particles such as dust, raindrops and fog, along with geometric transmission loss, atmospheric turbulence and pointing error are the main factors that severely degrade the link performance [109]. Atmospheric scattering can be modelled using frequency selective fading channel. In optics-based space communication, atmospheric turbulence can cause variations in the transmitted laser intensity and degrade the link. Atmospheric turbulence can be modelled using the log-normal and exponential model. The flicker in the laser can be modelled using the Gamma-Gamma distribution. Similarly, the distribution of the pointing error depends on multiple factors including the ratio of the equivalent beam radii, standard deviation of the pointing error offset, error function, beam width, beam emission angle and receiver radius.

B. GSO and NGSO SATCOM Modulation Schemes

A study of different modulation schemes including minimum-shift keying (MSK), serial MSK, BPSK, QPSK, $\pi/4$ -QPSK, and offset QPSK (OQPSK) for a SATCOM modem has shown that constant envelope modulation schemes (frequency-shift keying (FSK) and phase-shift keying (PSK)) with a band-limiting filter avoid the nonlinear behaviour at high power and suppress sidelobes [42]. But FSK schemes are not suitable for SATCOM due to low bandwidth efficiency. For SATCOM systems, the constant phase modulation (CPM) is preferable to PSK due to continuous phase transition which yields fewer sidelobes.

BPSK, QPSK, OQPSK, and MSK depict similar power efficiency. But parallel MSK requires phase and time synchronization and amplitude balancing. BPSK, while simple to implement, shows the worst performance in terms of bandwidth efficiency. On the other hand, MSK, with a complex implementation architecture, shows excellent immunity to noise and moderate bandwidth efficiency. $\pi/4$ -QPSK and OQPSK are better suited for SATCOM than QPSK due to:

- (i) smaller fluctuations in the spectrum;
- (ii) better immunity to nonlinear behaviour of high power amplifiers; and
- (iii) moderate complexity for implementation.

With new developments in the SATCOM modem in recent years, the multi-mode modem with modulation and channel coding may be used to gain power efficiency and in turn increase bandwidth efficiency. For small SDR-based SATCOM systems, adaptive coding and modulation schemes must be employed to take advantage of the time-variant SNR to achieve high throughput [99], [101], [110]–[114].

The digital video broadcast-S2 (DVB-S2) is a well known standard for SATCOM systems. The demapper algorithm for

decoding the modulated information in DVB-S2 has been discussed in [115]. A max-log algorithm that exploits the symmetry of the constellation points in PSK modulation schemes to yield linear complexity is used for demapping of 8PSK symbols. Based on a compare-select technique for log-likelihood ratio (LLR) computation, the Euclidean distance calculation for each bit is avoided since the constellation points exhibit symmetric LLR values.

A study on LEO satellite systems has reported a growing demand for higher resolution imagery from remote satellites [116]. But with a fixed, limited frequency band available this can lead to congestion in the satellite-ground link. A survey on small satellites using different modulation schemes such as $\pi/4$ -QPSK, OQPSK, and Gaussian minimum shift keying (GMSK) as well as different coding schemes such as convolutional 3/4 and Reed-Solomon (12,9), and (255,243) codes, has concluded that while small satellites are promising, they are handicapped by their size which imposes constraints on power and capacity [101]. Throughput can be increased by adopting desirable coding and modulation schemes with Rician channel modelling and log-normal fading models.

The Consultative Committee for Space Data Systems (CCSDS) has recommended GMSK and OQPSK with trellis coded modulation (TCM) with inherent convolutional coding 4D-8PSK-TCM for remote sensing applications. Transfer of high resolution (0.3m) imagery from a satellite to the ground terminal requires high data rate (1.5 Gbps). The commonly used communication system cannot meet this requirement. In [116], a flexible modulation scheme has been proposed with a simpler modulation and demodulation principle for remote sensing satellite systems. The system can achieve high data rates of 24 b/s/Hz with optimal SNR distance to Shannon bound. The proposed modulator comprises two state modulation, i.e., m -PSK with constant envelope amplitude A_1 and low order amplitude based modulation A_2 . For demodulation purposes, the second channel phase is irrelevant, hence can be fixed to achieve maximum orthogonality of signals between two channels. Furthermore, the authors derive the sub-channel energy and spectral efficiency of the proposed 16PSK + PAM4 modulation scheme.

In [117], the BER and frame error rate (FER) performance have been studied to validate a software-defined baseband (SDB) system with telemetry receiver and telecommand transmitter for LEO satellites. To validate the transmission capability of the SDB telecommand transmitter, a power spectrum characterization has been performed. It was found that the modulation index determined the allocation of power. Thus, for a power limited communication system, the trade-off between synchronization of the signal and achievable bit rate depends on the modulation index. For the uncoded BPSK modulation scheme, the performance of the telemetry receiver is within 1 dB of ideal performance. For coded BPSK, Reed Solomon (RS), CC, concatenated codes and LDPC codes were tested. The results depict that performance is within 1 dB for RS(255,223) and RS(255,239) with respect to the CCSDS green book performance curve [118]. The loss in coding gain is due to synchronization loss of the Costas loop. For CC ($k = 7$, rate = $\frac{1}{2}$), performance loss of 2.1 dB

TABLE VI: Comparison of SATCOM Modulation Schemes.

Modulation	BW Efficiency	BER Tolerance	Complexity
ASK	Not suitable due to nonlinear behaviour of HPA		
MSK	Moderate	Moderate	Nominal
GFSK	Moderate	Moderate	Reasonable
BPSK	Poor	Desirable	Nominal
QPSK	Better	Desirable	Reasonable
8PSK	Best	Poor	High
APSK	Best	Reasonable	Highest

is observed in BER and 0.7 dB in FER. The loss is due to reduction in transmitted power in $CC(7, \frac{1}{2})$ which results in synchronization error at low bit-to-noise ratio $\frac{E_b}{N_0}$. For concatenated codes, i.e., RS with convolutional decoder, BER and FER suffer degradation of 2 dB with respect to the CCSDS green book. The degradation in performance is due to reduced loop SNR creating synchronization issues. For LDPC codes, the implemented decoder is applicable to CCSDS specified standards [118]. However, C2 codes are used as they are specified for near earth missions. The performance of the decoder is comparable to the CCSDS LDPC decoder for LDPC codes [119]. For practical validation of the SDB system, 3 LEO satellites operating in S-band are considered. The performance analysis based on the BER and FER is difficult to calculate due to unknown a priori telemetry transmitted frames, unknown and varying carrier-to-noise ratio $\frac{C}{N_0}$ values, and satellite pass time. Hence, a rudimentary approach is used based on demodulated signal constellation and decoder errors to validate the FEC functionality. For the satellite with BPSK signal, 2 Mbps data rate and 15.77-19.51 dB SNR, no error was found using Viterbi and RS decoders. For the satellite with 235.294 kbps data rate, QPSK modulation scheme and 12.1-23 dB SNR, error free performance is observed using both decoders. Similar decoder behaviour is observed for the remaining satellite with 1.024 Mbps data rate with OQPSK and 12-14 dB SNR.

Laser communication can meet the high speed data transmission requirements for advanced SATCOM systems unlike traditional microwave links. More specifically, the data transmission of laser is $100\times$ more than microwave communication [109]. Moreover, laser communication is safer due to its high beam directivity. M-QAM modulation scheme with OFDM is used to evaluate the performance of the laser link, based on BER and transmitted power. OFDM based modulator and demodulator is designed that can generate data rate of 10 Gbps with 4-QAM and 1024-point inverse FFT. 512 sub-carriers for OFDM transmission are used with distributed feedback semiconductor laser light source. The spectral width of laser is 10 MHz. The link is evaluated for different atmospheric turbulence, and QAM order $M = 4, 16, 64$.

A comparison of different modulation schemes is presented in Table VI and recommendations are summarized in Table VII.

IV. SATCOM CHANNEL ENCODING

ITU-R has defined enhanced mobile broadband (eMBB), massive machine-type communication (mMTC) and ultra-reliable low-latency communication (URLLC) as the use cases

for 5G systems [120]. Using URLLC, SATCOM systems are expected to deliver latency < 1 ms and packet loss rate $< 10^{-5}$. The 3rd Generation Partnership Project (3GPP) has identified satellites as a game changer in the 5G ecosystem. Moreover, [121]–[123] have identified the non-terrestrial network (NTN) as a new feature of 5G networking systems. NTN 5G systems can:

- (i) ensure service continuity for mission critical applications as well as communications on the move applications,
- (ii) ensure service guarantee in unserved as well as under-served areas by supplementing the terrestrial network coverage, and
- (iii) offer service scalability due to the wide area satellite coverage, delivering better service quality to users [124].

In the following, we provide a detailed discussion on the state-of-the-art in SATCOM channel encoding for various block lengths and the related challenges.

A. Long vs Short Block-Length Channel Encoding

To achieve the reliability and low latency requirements for 5G communication systems and satellite constellations, channel coding plays a crucial role. While a number of channel coding techniques are available, tail-biting-CC (TB-CC), low-density parity-check (LDPC) codes and polar codes (PC) have been recommended for 5G communication systems, in particular for URLLC and eMBB scenarios.

Channel coding for long block-lengths has yielded impressive results. Modern schemes such as LDPC codes and turbo codes (TC) have shown exceptionally good performance for block lengths $> 2,000$ bits with simple decoding algorithms such as the iterative belief propagation (BP) [125]–[133].

On the other hand, channel coding for short block-lengths remains an open problem due to the conflicting requirements of reliability and low latency; powerful codes with long block-lengths offer reliability at the expense of latency while short block-length codes offer low latency but compromise reliability [19], [101], [133]–[139]. Classical techniques for code construction like density evolution and extrinsic information transfer (EXIT) charts fail to deliver codes with acceptable performance due to asymptotic properties of density evolution and the failure to properly model the iterative decoder for EXIT chart analysis [135]. The most viable candidates are short binary and non-binary LDPC codes, TC, (128,64) extended BCH codes, two-TB-CC, and PC using successive cancellation list (SCL).

B. LDPC Codes for Wireless Communication

LDPC codes circumvent the need for interleavers, trellis-based decoding algorithms and cancellation lists so have lower implementation complexity as compared to TC, TB-CC and PC [19], [127], [128], [135], [136], [138], [140]–[143]. Some examples of LDPC codes include the Gallager LDPC codes, finite-geometry LDPC codes, random LDPC codes [144], algebraic LDPC codes [145], protographic LDPC codes based on EXIT (PEXIT) chart [135] and incomplete geometry based LDPC codes [146].

TABLE VII: Modulation Schemes for SATCOM

Modulation	Recommendation
ASK	• Not recommended due to nonlinear behaviour of HPA (TWTA)
FSK	• Low bandwidth efficiency
PSK	• BPSK, QPSK, OQPSK, and MSK have the same power efficiency • BPSK is simplest to implement but has the poorest bandwidth efficiency and immunity • QPSK and OQPSK have the best bandwidth efficiency and moderate immunity and complexity • MPSK has the best immunity, moderate bandwidth efficiency but highest complexity • $\pi/4$-QPSK is suitable where differential detection is needed • BPSK has the best error tolerance but the worst transmission efficiency • 8PSK has the best transmission efficiency but the worst error tolerance

LDPC codes can be constructed by decomposing large matrices (blocks) into sub-matrices which exhibit a cyclic property, i.e., rows of sub-matrices are related via cyclic shifts, leading to the quasi-cyclic LDPC (QC-LDPC) codes. Some popular examples include the finite-geometry QC-LDPC codes [144], protographic QC-LDPC codes [136], [147] and QC-LDPC codes based on fundamental theorem of arithmetic [148]. Due to their structural properties, QC-LDPC codes are easier to implement as compared to other LDPC codes [105], [147]–[155].

LDPC codes based on a lower triangular matrix approximation of the parity check matrix H lead to a reduction in the encoder complexity [130]. The lower triangular form of H can be achieved rapidly using permutations of the columns or rows instead of the Gaussian elimination procedure. LDPC decoders implement a message passing algorithm between the sets of check-nodes with distinct variable nodes [131]. This not only reduces the number of iterations needed for convergence but also the memory requirement as compared to the flooding schedule and semi-serial schedule methods. A study on the BER performance of the LDPC, CC, and uncoded schemes for 16QAM and BPSK has shown that LDPC codes, with a lower complexity, outperform CC [127]. Under channel fading, [105] has discussed a multi-rate LDPC encoder where the parity check matrix H is converted to a lower triangular matrix by applying the RU algorithm. The shift-register-adder-accumulator (SRAA) reduces the encoder complexity for QC-LDPC codes as compared to TC. In the Chinese DMB-T transmission, the generator matrix G can be used in the systematic form instead of the H matrix, admitting parallel processing in the SRAA which further improves the encoding rate.

Finite geometry-based LDPC codes exhibit the cyclic property and can be decomposed into a large class of quasi-cyclic and cyclic finite geometry LDPC codes using the roots of the generator polynomial of the mother cyclic code, permutation factors and masking operations while satisfying the row-column constraint on the H matrix [150]. The cyclic and quasi-cyclic codes have an advantage for hardware implementation as they enjoy linear complexity and can be implemented using shift registers. Construction of the H matrix from the base matrix comprising identity and zero sub-matrices has shown that in QC-LDPC codes the check node and the variable node share strictly only one edge making parallel update of likelihoods possible [142]. This has a huge advantage for high throughput and low latency requirements beyond 4G and 5G communication.

An alternative approach for constructing QC-LDPC codes is using masking with the successive cycle elimination method [156]. This method uses a dispersion-based circulant permutation matrix (CPM) where the undesired CPM in the H matrix is replaced by a zero matrix until a target masking termination condition is satisfied that achieves the desired code. Another approach for constructing QC-LDPC codes is using base graphs (BG) or protographs which are replicated to achieve larger graphs resulting in two different classes of protographic-based QC-LDPC codes, namely BG1 and BG2 [136].

A survey on decoding schedules for LDPC-CC using application specific integrated circuit (ASIC) and field programmable gate array (FPGA) implementations has shown that for optimized algorithms and in particular, the binary memoryless symmetric channel (AWGN and binary erasure) and in general for all multi-user channels, LDPC-CC is capable of achieving the decoding threshold of maximum a posteriori probability (MAP) via the BP message passing algorithm [132]. The implementation of the sum-product algorithm (SPA) for BP has high complexity due to the nonlinear $\tanh$ operations and so is not recommended for very large-scale integration (VLSI) implementation. A computationally cheaper alternative is the min-sum approximation and its variants that can yield a reduction in the hardware complexity of up to 20%. Moreover, the on-demand variable node activation (OVA) with layered based scheduling approach can overcome the inefficiency of the iterative message update schedule. To achieve a throughput of Gbps, pipelined architectures for FPGA and ASIC implementations are used but these suffer from extreme latency. Thus, are unsuitable for real time applications. In order to overcome latency, single processor LDPC-CC decoders have been studied but these are unable to achieve the desirable decoding throughput. To achieve: (i) Gbps throughput, (ii) low latency in the decoder, and (iii) low memory requirement, both schedule and node level optimization is needed.

In a construction based on the PEXIT method and selective avoidance of cycles (ACE), the appropriate correction properties of protograph are selected based on the PEXIT analysis and the edges of the protographs are permuted based on the ACE method [141]. A comparison with the LDPC codes (in specification CCSDS 231.1-O-1) has shown that this approach outperforms LDPC codes with a 30% reduction in decoding complexity. Moreover, storage of only the H matrix is required, unlike LDPC codes which require storage of the G matrix.

A finite geometry-based QC-LDPC code is constructed

using an auxiliary matrix to facilitate identification and elimination of 6-cycles of the block sub-matrices obtained from EG-LDPC codes [153]. Based on the randomization and circulant shift method, regular QC-LDPC codes are constructed using the XOR operation, uniform distribution of non-zero elements and pairing of rows and columns [152]. A systematic design of the protograph-based QC-LDPC codes can also be achieved by elimination of the trapping sets in a protograph [147].

C. Channel Coding Schemes for Next Generation Systems

In [135], the performance of various channel coding schemes for block length $n = 128$ and code dimension $k = 64$ has been presented in terms of coding gain and decoding complexity. This is the shortest code for satellite telecommand link standardized by CCSDS. More specifically, a comparison is shown for 128/64 binary protograph-based LDPC codes, AR3A LDPC codes, ARJA LDPC codes, (3, 6) LDPC regular codes, CCSDS LDPC codes, TC, BCH codes, TB-CC and PC. For $\text{CER} = 10^{-4}$, the coding gain of CCSDS LDPC codes is low relative to that of AR3A LDPC codes, (3, 6) regular LDPC codes, and ARJA LDPC codes. However, for the satellite telecommand link, error detection is more critical than coding gain and so CCSDS LDPC codes, which achieve low error rate, $\text{CER} = 10^{-6}$ compared to other LDPC codes, are preferred. Also, for $\text{CER} = 10^{-4}$, TC outperform binary LDPC codes but are outperformed by BCH codes. PC with cyclic redundancy check (CRC) outperform all other codes for $\text{CER} = 10^{-4}$ and $\text{CER} = 10^{-6}$. TB-CC codes with order statistic decoding (OSD) achieve the performance of BCH and LDPC codes. With respect to decoding complexity, the performance of iterative decoding is satisfactory for binary LDPC codes and TC with the advantage of hardware implementation. For non-binary LDPC codes, the FFT-based decoders are $\approx 64\times$ more complex than the decoders for CCSDS LDPC codes. However, the complexity of the FFT-based decoder can be reduced using sub-optimal check node update rules. OSD and non-binary BP decoding algorithms have similar complexity for short block-lengths but OSD may be impractical for long block-lengths.

For URLLC, mMTC, and eMBB scenarios in 5G with block size in the range [40, 1000] bits, a comparison of 3GPP LTE standard TC, LDPC codes, PC, and CC based on the BER performance for a given code rate and modulation technique has shown that selection of the coding scheme depends on the block length [138]. For short block-length (40 bits) using QPSK and 16 QAM with code rate $1/3$ and $\text{BLER} = 10^{-6}$, PC and LDPC codes exhibit the best performance. But for the lower code rate of $1/12$, PC with SCL decoding outperform all other schemes. PC with SCL depict 0.25-2 dB improved margin with respect to other coding schemes. We note that for LDPC codes, the CRC is not required since the syndrome calculation is sufficient to validate the decoded codeword. However, the performance of LDPC codes is limited by the shortest girth, as in the case of progressive edge-growth (PEG). On the other hand, for medium-block length (200 bits) and code rates $1/3$, $1/12$, TC outperform LDPC, PC and CC. LDPC shows the second best performance with 0.5 dB loss.

With respect to decoding complexity, LDPC codes depict better performance with the min-sum algorithm than with SPA. The soft input Viterbi algorithm for code rate $R = 1/3$, length = 9 and generator polynomials 557_{oct}, 663_{oct}, and 711_{oct} gives better performance for terminated CC. The inclusion of SCL results in a noticeable improvement in the performance of PC.

Under reliability and high throughput requirements, CC with bitwise map decoder based on the Bahl-Cocke-Jelinek-Raviv (BCJR) algorithm, PC with successive cancellation, and QC-LDPC codes with SPA for decoding, have shown similar BER performance for BPSK in AWGN channel [143]. We note that unlike CC, the decoder implementation for TC also requires an interleaver. The decoder uses the quadratic permutation polynomials (QPP) with two soft input and soft output (SISO) decoders. For QC-LDPC codes and repeat-accumulate codes, SPA gives satisfactory results.

In 3G/4G communication, the primary coding scheme is TC, obtained from a permutation of CC with trellis-based one-dimensional graphical structure decoding. However, for high throughput requirements of 5G communication, TC are replaced by LDPC codes which support multiple block sizes and rates while observing strict performance parameters with reasonable complexity. LDPC codes have nice structural properties that are amenable to parallelization in hardware as compared to TC. Moreover, LDPC codes are conveniently defined using the parity check matrix H and Tanner graphs. The construction of QC-LDPC codes can be achieved by replicating the BG with permutations. The latest LDPC codes for 5G communication are essentially a concatenation of the LDPC codes with a low-density generator matrix (LDGM). During the check node update procedure the updated variable node may not be available so it is not efficient for decoding. However, in quasi-row orthogonality construction, the orthogonality between consecutive layers avoids dependency and so is efficient for decoding. Considering BG1 with information bits ($500 \leq K \leq 8448$) and rate ($1/3 \leq R \leq 8/9$) and BG2 with information bits ($40 \leq K \leq 2560$) and rate ($1/5 \leq r \leq 2/3$), BG2-based LDPC codes have superior BER performance than TC.

QC-LDPC codes for a given expansion factor (or lifting degree) and fundamental theorem of arithmetic produce three classes of QC-LDPC codes with girth of at least 6 [148]. Assuming an AWGN channel with fast fading and quasi-static flat Rayleigh fading for binary input channel, the word error rate (WER) performance of LDPC, Golay, Reed-Muller, random codes, PC, TC, and BCH codes for (very) short block-sizes (i.e., 256, 32, and 20 bits) using OSD depicts that in the absence of CRC, BCH outperforms the other schemes [133]. However, with the inclusion of CRC and joint decoding, all schemes depict very similar performance that is close to the optimal lower bound. Moreover, optimization of OSD based on probabilistic necessary condition, probabilistic sufficient condition, reference re-encoding scheme, pre-processing rules and multiple biases diversity scheme can reduce the complexity making it feasible for short length decoding operations.

In [134], the performance of various codes for short block-length ([50, 1024] bits) has been assessed with respect to

finite-length performance bounds, i.e., converse bounds based on the metaconverse (MC) theorem and achievability bounds based on the random coding unit (RCU) bound. A performance comparison of Golay codes, BCH codes, TB-CC, CRC concatenated TB-CC, binary and non-binary TC and LDPC codes (PEXIT method, ARA-LDPC, ARJA-LDPC, H-ARQ and QC-LDPC (BG2)) and PC, has shown that for short block-lengths (< 128 bits), TC and LDPC codes are outperformed by Golay, BCH and TB-CC codes. However, for moderate lengths in the range $[128, 512]$ bits, the performance is comparable. The CRC concatenated TB-CC results in better error detection and correction performance as compared to simple TB-CC. For block length of $(128, 64)$, and $\text{CER} = 10^{-6}$, LDPC codes based on BG2 for 5G NR, outperform other LDPC schemes.

A comparison of PC, TC and LDPC codes with respect to features and applications of 5G communication has shown that large decoding delay and multiple iterations for TC render it difficult to meet the network requirements of high speed and low latency [19]. Moreover, systems using multivariate LDPC code have better bandwidth utilization, and deliver better performance for moderate and short code-length than PC. The concatenation of LDPC code and PC (LDPC-Polar or Polar-LDPC) may yield promising results for 5G-communication systems. For URLLC with optimal decoding, a comparison of TC, PC, LDPC, BCH, and CC codes has shown that due to the minimum Hamming distance, BCH codes outperform all other schemes in terms of reliability [137]. Moreover, for $\text{CER} = 10^{-4}$, and with reasonable complexity, PC using SCL showed only a 0.5 dB gap to the normal approximation.

A systematic procedure for construction of hybrid quasi-cyclic LDPC (HQC-LDPC) codes for short block lengths along with BER performance comparison with existing LDPC codes is given [157] for M -PSK modulation scheme and different code rates. The proposed method exploits Euclidean geometry and circulant decomposition resulting in a QC-LDPC matrix that avoids shorter girths for iterative decoding cycles. The construction of HQC-LDPC matrix is systematic in nature with a matrix-based construction approach. The constructed HQC-LDPC matrix has the characteristics of EG, circulant decomposition and random parity-check matrix. The BER performance of HQC-LDPC is discussed for M -PSK where $M = 2, 4, 8$, and for code rate of $\frac{1}{2}$ and $\frac{3}{4}$. Table VIII and Table IX show a comparison summarizing the BER performance of HQC-LDPC codes, QC-LDPC codes, e.g., BG1 and BG2 [136], QC-LDPC using dispersion and masking [156], algebraic QC-LDPC, QC-LDPC using permutation based masking matrix and protograph QC LDPC [148].

V. SOFTWARE-DEFINED RADIO BASED SATCOM SYSTEMS

A software-defined radio (SDR) is a hardware device which implements signal processing methods such as digital conversion, upsampling, modulation, filtering, and detection using software-based algorithms. An SDR has the capability to perform partial or full physical layer function of a radio using modifiable software and programmable hardware [16], [158]. Once programmed, unlike a normal radio, an SDR has the

flexibility to reprogram its working architecture to evolve with advanced communication protocols, channel assignment policies and wireless functionalities [16]. The signal processing algorithms can be implemented using tools such as WARP, NI LabVIEW, NI USRP, GNU Radio, IMEC SDR, Lyrtech SDR Platform, etc. Recent developments focus on reconfigurable SDR System-on-Chip ground receivers with the ultimate aim of Satellite-on-Chip [159], [160]. We begin with a brief review of SDR for wireless transmission and then turn to SDR-based SATCOM systems.

In [161], the authors have analysed the JPEG file transfer over a wireless channel using the GNU Radio, a software development platform for Universal Software Radio Peripheral (USRP) devices. The transfer protocols are implemented using the GNU Radio libraries. The differential binary-PSK scheme with a bit rate of 500 kbps and packet size of 1500 bytes was used for transmitting the file. The received file was found to be distorted due to packet loss during the transmission so this approach was neither robust against channel interference and duplication nor robust for packet recovery. The USRP-based wireless transmission protocol implementing buffering and leaky bucket algorithm showed significant improvement in transmission error and environmental interference problems [162]. The setup includes three nodes with a transmitter, receiver, and repeater. A Python/C++ environment is used for the development platform. The simplified wrapper and interface generator is used to interface Python and C++. The presence of a buffer helps to receive the correct transferred file size. Furthermore, an identification for the sending and receiving modules was added for multi-node communication in addition to time delay between transmission of successive packets. The buffering and leaky bucket algorithm control the flow and time for sending several packets. Thus, mitigating distortion and duplication issues arising in transmission over a wireless channel.

SDR prototyping platforms over heterogeneous and multicore processors overcome the challenges associated with central processing unit (CPU)-related computational complexity [163]. The 20 megasamples per second (Msps) I-Q data are upsampled to 400 Msps and transmitted at RF band. At the SDR receiver, signals are converted to baseband I-Q components using a direct conversion receiver (DCR). The digitization of a signal is performed at 100 Msps using a 14-bit analog-to-digital converter (ADC) with real-time processing handled using an FPGA.

The existing literature on SDR for SATCOM has discussed the challenges as well as advantages of small satellites [99], [102], [110]–[112], [114], [126], [159], [162], [164]–[170]. Real-time implementation issues in terms of computational capability and processing delay of an SDR have also been studied [159], [162], [163], [169]–[172]. These studies have concluded that multicore and heterogeneous processors can overcome real-time challenges.

But an SDR implementation of the multiple access control (MAC)/physical layer of wireless/SATCOM networking has not received much attention. The virtualization of the MAC/physical layer can be implemented using an SDR for space related applications [173]. In the following, we review

TABLE VIII: BER Performance of HQC-LDPC for 1/2 Code Rate.

FEC Type/ Reference	Decoding Algorithm	Code Rate	Block Length	Modulation/ FEC Type	BNR vs BER (10^{-2})	BNR vs Error Floor (EF)
HQC-LDPC	SPA	0.5	128/256	BPSK	1.55 dB	3 dB
				QPSK	1.7 dB	3.5 dB
				8PSK	3.9 dB	5.5 dB
Random LDPC	SPA	0.5	128/256	BPSK	1.75 dB	4.5 dB
				QPSK	2.2 dB	5 dB
				8PSK	5.4 dB	7 dB
[136]	SPA for BG1 & BG2	0.5	128/256	BG1 & BG2 Protographic QC-LDPC	2.5 dB (BLER)	4 dB (10^{-4})
[156]	SPA	0.5	288/576	QC-LDPC (Dispersion & Masking)	2.75 dB (WER 10^{-3})	4 dB (10^{-4})
[148]	SPA	0.5	1024/2048	Algebraic QC-LDPC	1.8 (BER 10^{-3}) QC-LDPC	3 (EF)
				QC-LDPC using permutation & masking	1.8 (BER 10^{-3})	3 (EF)
			525/1050	PEG QC-LDPC	2.05 (BER 10^{-3})	3 (EF)
				QC-LDPC using permutation & masking	2.05 (BER 10^{-3})	3 (EF)

TABLE IX: BER Performance of HQC-LDPC for 3/4 Code Rate.

FEC Type/ Reference	Decoding Algorithm	Code Rate	Block Length	Modulation/ FEC Type	BNR vs BER (10^{-2})	BNR vs Error Floor (EF)
HQC-LDPC [157]	SPA	0.75	192/256	BPSK	3.2 dB	4.5 dB
				QPSK	3.25 dB	5 dB
				8PSK	6.4 dB	9 dB
Random LDPC	SPA	0.75	192/256	BPSK	3 dB	5 dB
				QPSK	3 dB	5.5 dB
				8PSK	6.15 dB	9 dB
[136]	SPA for BG1 & BG2	0.75	500/667	BG1 & BG2 Protographic QC-LDPC	4.9 dB (BLER 10^{-2})	5.5 dB (10^{-4})
[156]	SPA	0.75	432/576	QC-LDPC (Dispersion & Masking)	3.75 dB (WER 10^{-3})	4.75 dB (EF)

developments in SDR for multiple orbit SATCOM systems with particular emphasis on processing and transmission capabilities and limitations of SDR-based modem implementation.

An SDR can implement adaptive and multi-standard communication systems due to its ability to perform digital signal processing on devices (i.e., general purpose processors and FPGAs). The development platform of an SDR is highly flexible allowing modification using an integrated development environment (IDE) [160]. However, the practical implementation of these devices for SATCOM systems is still limited. Recently, there is considerable interest in communications on the move environment driven by small satellite systems so there is considerable scope for SDR implementation for such systems. One practical example is SDR-based testbeds for GEO SATCOM systems. These testbeds can be used to study interference mitigation in multibeam scenarios. Another such practical example involves the testing platforms for small satellites, in particular, the CubeSat system. Multiple companies (GomSpace, NanoAvionics, Alén Space, ISISpace) have built SDR payloads for the CubeSat platform based on system-on-chip using FPGAs, complex programmable logic devices and processing units. The DVB-S2 compliant GomSpace platform is one such example [30].

Channel coding is an important component of SATCOM systems, therefore it has received considerable attention for SDR-based applications [174], [175]. DVB-S2 is one of the most popular protocols for SATCOM systems and has been investigated for transceiver design in [176], [177]. One of the popular error correcting codes used in DVB-S2 and DVB-S2X standards are the LDPC codes which have shown BER performance close to the Shannon limit with low complexity and low latency. The challenges relating to practical implementation of LDPC codes have been overcome by the flexibility of the SDR approach. Simulation of small satellite constellation via an SDR is an emerging research area. The reconfiguration capability of SDR is useful for waveform design and to ascertain synchronization parameters [178].

A. SDR for the GSO SATCOM System

Only a few studies have been conducted on SDR-based testbeds for the analysis of SATCOM systems for both communication on the pause and communication on the move applications [99], [102], [114], [126], [166], [168]–[170]. Most of the work has focused on NGSO, i.e., small LEO satellites [38]–[40], [99], [101], [113], [114], [126], [133], [137]–[139],

[166], [168], [169]. Popular FPGA platforms used in these SDRs are the Zync 7000 hybrid ARM/FPGA and KINTEX7-410T FPGA [30], [117].

In [172], an SDR-based system for the GEO-orbit satellite has been developed. The payload signal is the DVB-S signal using the QPSK modulation scheme and the ranging signal is BPSK modulated with PN sequence with a symbol rate of 2.048 Msps and 70 MHz bandwidth. The study has shown that the use of PN techniques for ranging measurements allows the same bandwidth for both ranging and payload signal with negligible impairment to the latter. For the geostationary orbit, DVB-S and ranging signal can be treated as additive noise with respect to each other. Moreover, for a measurement time of 1 s and root mean square error of about 25 cm, the relative power of the payload to the ranging signals is measured to be as small as -18 dB.

In [159], an SDR-based distributed satellite system has been studied where the link is evaluated using BPSK with concatenated codes comprising Viterbi (1/2) and Reed Solomon (RS) codes (160,128) for symbol rates of 1, 2.4, 4.8, 9.6, and 19.2k. A performance comparison using FPGA and CPU for the distributed satellite system has shown that an embedded system architecture is suitable for the earth stations favouring transition in communication from a single satellite to multi-satellites. Distributed and mobile ground terminals can be deployed to enhance the access time. Moreover, an embedded design for future SDR-based distributed SATCOM systems is required to compensate for the development requirements.

In [179], SDR emulators for channel modelling of GEO satellites have been developed using USRP and LabVIEW IDE. The emulator models the channel using the thermal noise characteristics, phase noise simulating rotation of symbols, propagation delay encountered during transmission and furthermore, processes a 1.6 MBaud signal stream. Interference mitigation for GEO satellite systems has been studied based on acquisition, measurement and mitigation of RF interference. The infrastructure has been developed using several devices and a hybrid approach; the payload emulator is based on BEECube's BEE4, the channel emulator is based on the single-link satellite channel emulator RT Logic T400CS, and the transceiver is based on the platforms provided by Nutaq Inc. Interference mitigation schemes can be found in [180]–[182]. In an SDR based wide-band radio multi-system avionics architecture for the satellite link, [181] has used adaptive coding and modulation to establish the communication link. In order to reduce the interference generated from multiple sources in the wireless communication system, the so-called multi-linear subspace estimation and projection algorithm has been proposed that outperforms projection based radio frequency interference detection algorithms at the cost of a slightly higher complexity [180], [182]. Multi-beam capability is important for high throughput systems. The SERENADE NI USRP based on an end-to-end multi-beam SATCOM emulator has been used to evaluate precoding and interference mitigation strategies [183]–[185]. The transponder for LEO to GEO orbit is emulated for forward link via DVB-S2X emulators with multi-beam capability and multiple user terminals.

In [186], an SDR-based receiver frontend architecture has

been designed for broadband SATCOM systems. A multi-core SDR architecture offers flexibility and low production cost for GNSS receivers [187]. Besides SDR-based modules for source coding, modulation, channel coding and multiplexing, SoftSpace based MAC/physical-layer function virtualization has been designed and developed on FPGA and signal processing boards in space forwarding devices [188]. The interoperability of these devices can be promoted by abstracting the MAC/physical layer functions via reprogrammable and independent SDR technology having signal processing and resource sharing capabilities [173]. The use of SDR in these devices can enhance the adaptive features of the underlying communication technology, reconfiguration of the physical layer parameters and optimization of resource utilization. Similarly, in heterogeneous network topology for SATCOM systems, SDR can provide the capability of far-end upgrades, reconfiguration of implemented radios and multi-mode operations.

In [189], an SDR-based prototype design of a SATCOM modem for geostationary satellites with a bent-pipe satellite transponder and LABVIEW-based base-band signal generation and processing architecture has been discussed. The proposed design accommodates multiple provisions such as variable length message signal, variable packet structure, option of CRC, RS codes for short block-lengths at various rates, DSSS with various modulation schemes including BPSK, QPSK and 8PSK at various transmission rates and various reliability requirements. In addition, the proposed design incorporates anti-jamming capability and user anonymity along with self-reliance. The path loss has been modelled using AWGN and a delay of 250 ms has been incorporated to model the propagation delay in the SATCOM channel. USRP-X310 is used for the hardware implementation. The prototype is a testing platform for communication in both static and dynamic GEO, MEO and LEO environments.

B. SDR for the Non-GSO SATCOM System

The feasibility of a low cost SDR prototype for the ultrahigh frequency (UHF) and very high frequency (VHF) bands using commercially available off-the-shelf (COTS) components has been investigated in [110]. Doppler shift was identified as one of the main challenges in the communication between the ground station and LEO satellites. Communication with the LEO space segment requires software controlled (Ham Radio Deluxe) tracking antennas that can be steered in real-time to the predicted satellite location and can automatically adjust to compensate for the Doppler shift. A low cost solution can have some performance limitations in terms of signal processing, multi-frequency communication and multi-mission capabilities.

In [111], an SDR-based ground station transceiver (USRP N210) has been studied which incorporates the design of the mission control center for the European Student Earth Orbiter (ESEO) project by developing an SDR-based earth station segment. The design is based on amateur radio frequencies and AX.25 communication protocol and can be used for downlink and uplink transmissions. All digital functions are

performed on an FPGA board with modulation schemes that are UHF-band compliant pulse coded modulation with 0.5 bandwidth-time factor for Gaussian frequency-shift keying (GFSK) having non-return to zero characteristics. The ground station can support inter-satellite communication at UHF-band in the LEO.

Simulations of communication between multiple satellites orbiting in LEO using DS-CDMA, Rayleigh fading, and AWGN channels using QPSK and BPSK with CC have shown that an adaptive and reconfigurable communication system for the Inter Satellite Link (ISL) can achieve high data rate [99]. The design parameters include orbital velocity ~ 3 km/s, with distance between neighbouring satellites < 3 km, transmission range 10-20 km, data rate 3-8 Mbps, and transmission delay 150-300 ms. The challenge for small satellites is the limited available power. For transmission power 0.5-3 W and S-band, it is 1-2.5 W for 1U CubeSat and maximum 2-5 W for 2U CubeSat. The SATCOM channel is simulated using AWGN and Rayleigh fading models. In order to access the channel for inter-satellite communication, collision free schemes such as FDMA, TDMA, and CDMA, and contention based protocols such as ALOHA, carrier sense multiple access (CSMA), busy tone multiple access (BTMA), and idle signal multiple access (ISMA) may be used. However, contention based protocols are not suitable for real-time or near real-time applications. DS-CDMA due to its multiple user access with independent, uncorrelated pseudorandom codes can overcome the collision issue in multi-satellite communication. This is in contrast to FDMA and TDMA which require separate bandwidth and strict timing synchronization for multiple data rates.

A software-defined transmitter for a cube satellite based on high-end FPGA enhances the signal processing and flexible structure capability [167]. A performance comparison of BFSK and GMSK has shown that BFSK has the advantage of demodulation, low bandwidth, and higher efficiency at 9.6k baud rate. On the other hand, GMSK depicts good spectral efficiency due to the constant envelope feature and shows resilience against the amplitude base noise with moderate receiver complexity. Moreover, GMSK can be used with non-linear amplifiers providing low-cost solution and highly efficient C-type amplifiers and is suitable for both non-coherent and coherent detection systems. A model of a nano-SATCOM system based on the GNU Radio, HACK RF, and Raspberry Pi platform with GMSK, is a low-cost solution that is limited by amateur UHF RF and the size of the communication device [169].

The evolution in payload, telecommunication architecture, communication, and network protocols, all indicate to an increased use of small satellites for telecommunication services [126]. The utilization of higher frequency bands, optical communication, and advancements in protocols and architectures are some emerging trends in the small satellite field. Considering payload evolution, we find a focus on frequency band and antenna evolution, laser communication terminals for small satellites, and SDR payloads. The use of W-band (75-110 GHz) using horn antennas for small satellites is a modern trend. Similarly, the use of free space optical communication which can provide higher data rates at lower weight and power

consumption (10 Gbps within an order of 5kg weight and 50 W power).

For small LEO SATCOM systems, SDR-based adaptive communication using 16-amplitude phase shift keying (APSK), 8-PSK, QPSK, and BPSK schemes with coding rates $7/8, 5/4, 3/4, 2/3$, and $1/2$ using CC has shown that during an orbit the distance between the space and earth terminals varies resulting in changes in the SNR [113]. For maximum channel utilization, an adaptive modulation scheme can be used with high modulation like 16APSK when maximum SNR is available and low modulation BPSK when the SNR is critical with a feedback channel to change the transmission mode. However, incorrect frames observed may be due to the presence of noise, synchronization loss and rapidly changing Doppler shift. These issues can be eliminated employing additional channel coding and decoding, improvements in the synchronization algorithm at the receiver side, and implementation of good carrier tracking algorithms along with the use of short frames. Note that the use of these methods can enhance the transmission rate by about $2\times$.

A survey on communication and modulation protocols for small satellites has shown that due to the growing demand for small satellite missions, the traditional point-to-point communication protocols such as AX.25 reach their limit and novel, more efficient protocols are needed [114]. A low-cost SDR solution can accommodate the adaptive nature of communication in terms of modulation parameters and error coding capabilities with respect to the given channel conditions. Moreover, the NGHam packet architecture with RS error-correcting codes may be used to overcome the bottleneck. In the communication link for a LEO satellite, a carrier synchronization algorithm using non-linear operations can recover the LEO carrier [102]. Addressing the limitations of short visibility of LEO space segments, multiple ground stations at geographically diverse locations can enhance the availability time. However, this requires considerable ground infrastructure. An alternative approach based on adaptive modulation (MPSK/M-APSK) can be used to enhance the communication data rate. However, the Doppler shift is the main challenge for adaptive modulation schemes. A 1 kHzps change of Doppler shift demands a synchronization algorithm; otherwise, the advantages of high SNR cannot be achieved. A second-order type-2 phase lock loop (PLL) can track the carrier. In addition, a frequency correction block is required to remove the frequency shifts using FPGA devices without the assistance of a frequency locked loop.

Spectrum reuse and dynamic spectrum access are important considerations for next generation networks to achieve the goal of high-speed SATCOM [28]. The available satellite spectrum can be reused in a much more efficient way by deploying multi-beam, multi-band heterogeneous satellite transmissions and multi-orbit satellite systems. SDRs can be deployed for multi-beam satellite communication systems where interference between beam-cells can be mitigated via software control. Similarly, the heterogeneous nature of multi-band transmissions can be achieved using software switching of operating frequencies. For multi-orbit satellite systems, SDRs can be used to obtain GEO and non-GEO operational capabilities at

the software level.

In [190], an SDR-based LEO satellite receiver has been designed that can process data non-causally, implement MF for demodulation, and apply coarse Doppler compensation for full Doppler search. The MF enables the proposed SDR design to adapt to demodulate multiple modulation schemes such as FSK, MSK, GMSK and BPSK. The MF can be designed by curve fitting the received signal. The computational cost for the Doppler search is high so the NVIDIA Quadro K620 GPUs are used to reduce the processing time.

In order to extract the Doppler measurements for LEO multi-constellation systems, an adaptive SDR-based receiver has been designed and developed in [191]. The signals in LEO satellites have a higher SNR as compared to GEO and MEO satellite signals which is useful for positioning systems. However, communication is limited by satellite visibility and signal coverage. A multi-constellation positioning system can overcome these limitations. The authors have proposed the following procedure:

- (i) detect available satellites by using Welch's power spectral density,
- (ii) PLL-based tracking of Doppler frequency for available satellites, and
- (iii) select positioning measurements using switching filters.

Extended Kalman filtering (EKF) is used in receivers for tracking Iridium NEXT and ORBCOMM satellites. The ORBCOMM satellites operate in the VHF band with 137-138 MHz bandwidth with QPSK modulation and 4.8 kbps downlink data rate in a continuous mode of transmission. The Iridium NEXT constellation operates in 1616-1626.5 MHz with QPSK in burst mode. The adaptive design for multi-constellation receiver exploits the different modes of transmission, burst based simplex signal for Iridium-NEXT and continuous signal based transmission for ORBCOMM satellites [192], [193]. In order to support these multi-constellation, a switching mode for the EKF is designed which determines the available satellite based on the type and dimension of the measurement matrix. To test the performance of the receiver, bladeRF is used with low-noise amplifiers for Iridium NEXT satellite channel and RF bandpass filter for the ORBCOMM satellites. Considering the static mode, for ring alert burst mode the positioning error is reduced from 765 m to 213 m and for messaging burst modes it is reduced to 132 m resulting in a 72% improvement.

In [117], an SDR-based baseband system for satellite telemetry receiver and telecommand transmitter has been designed using the GNU Radio platform. The system has been validated using low cost SDR platform (USRP X310), and mission qualified satellite emulator in a lab environment and infield test were performed on orbiting satellites. The telecommand transmitter consist of three main units: (i) mission console centre (MCC) data transfer interface, (ii) FEC and frame generation block, and (iii) bit modulation block. Similarly, the telemetry receiver consist of three main blocks: (i) a bit synchronization block, (ii) FEC and frame synchronization, and (iii) MCC data delivery interface. The hardware test bed is based on the USRP X310 which is interfaced with the GNU Radio running on a Linux platform

with an Ethernet connection. In order to model the satellite channel, telecommand transmitter incorporates AWGN unit, and creates I/Q imbalance, frequency instability and symbol rate offset. It also incorporates the Doppler shift by introducing frequency sweeping. For validation purposes an intermediate frequency interface with parabolic a dish antenna configured to receive S-band signals has been used. The channel model for validation incorporates the amplifier non-linearities, delay and spread caused by the multiplexer, timing offsets including the Doppler shift, and multiple impairments, e.g., SDR RF distortion, phase noise, and DC offset.

In [194], the SDR is used to design a packet communication transceiver for the CubeSat and small LEO satellite systems. GNU Radio with Python is used as the software development platform for Raspberry Pi devices with channel access protocol AX.25. LEO satellites experience variations in the line of sight distance due to the relative motion between the space segment and the ground station resulting in different free space loss during the availability of the satellite link. This can be exploited to achieve an increased dynamic data rate based on the available link-budget [195]. Adaptive data rates, e.g., 2.4, 4.8, and 9.6 kbps with GMSK modulation are used for communication. Signalling packets are used to ascertain the feedback for the channel and to determine the corresponding data rates. The loss of feedback results in loss of the complete data packet due to the mismatch in the data rates. The transceivers require 7 ports with dedicated tasks and data rate with an organized timeline for collision free transmissions. The transmission simulations for multiple data rates at multiple SNRs depict that at higher data rates, the output power must be increased to account for the channel noise. Moreover, a 3 dB increase in the SNR doubles the system data rate. In the simulations, a 0.29 s packet transmission time was observed for 9.6 kbps, 0.57 s for 4.8 kbps and 1.13 s for 2.4 kbps. This resulted in a patch size of 50 for the proposed design in a single transmission. The timeout for reading the packets for the desirable packet success rate was observed to be 300 ms.

VI. FUTURE DIRECTIONS AND OPEN PROBLEMS

SATCOM systems can play an important role in achieving the objective of advanced wireless communication systems (i.e., 5G, 6G and beyond). These systems are being extensively explored in integrated-satellite terrestrial networks for 5G and 6G networks [196]. The main challenges are synchronization issues, resource allocation, channel modelling, trajectory optimization, interference issues, and new communication protocols for integrated systems [23], [197], [198]. Machine learning techniques for satellite applications, i.e., satellite operations, weather monitoring, earth observations and sensor fusion for navigation have the potential to enhance signal processing and resource allocation and furthermore, address the complexity issues in satellite networks [30]. In this section, we discuss future directions with respect to channel access methodologies, modulation schemes, channel encoding schemes, and SDR based applications and identify open problems.

A. Channel Access Methodologies

Advanced wireless networks aim to support high throughput (20-100 Gbps), better spectral efficiency (30 bits/Hz for downlink and 15 bits/Hz for uplink), optimum power consumption, hybrid systems (satellite and terrestrial systems), low latency (≤ 1 ms), communication reliability and high mobility (500-1000 km/h) [18], [24], [196]. Advanced channel access methodologies such as OFDMA, NOMA, WDMA, SDMA and rate splitting multiple access (RSMA) can play a pivotal role in achieving the above aims by enhancing flexibility and efficient use of the available spectrum.

OFDMA is a potential multiplexing scheme for future wireless communication systems [24]. It encounters multipath fading issues resulting in ISI for wireless and satellite communication. The addition of a CP can reduce ISI at the cost of transmission overhead. In addition, there is a power overhead, reducing the power efficiency of the system. SFBC-OFDM and NCC-OFDMA are variants of OFDMA that can resolve the multipath fading issue [60], [61]. OFDM also suffers from effective side lobe levels resulting in out-of-band (OOB) radiations (that are high amplitudes outside the required spectrum) reducing the spectral efficiency [199]. Multiple variants of OFDM such as F-OFDMA, GFDM, and TIBWB-OFDM have been considered for 5G and beyond wireless communication network [24], [67], [200]–[203]. SP-OFDM can provide secure transmission in the presence of a jammer [78]. F-OFDMA can potentially reduce the OOB radiations using spectrum shaping filters at the cost of complexity and interferences [62]–[64]. GFDM can enhance spectral efficiency due to requirement of only one CP at the cost of ISI, ICI, PAPR, and implementation complexity [18], [67]. To address these issues and to make it suitable for future systems, multiple equalization and interference cancellation techniques with advanced architecture have been proposed [66], [67], [67]–[73]. TIBWB-OFDM is an attractive choice for enhancing higher spectral and power efficiency [74]–[76]. However, further research is required to investigate suitability for future wireless communication systems [77].

With the advancements in satellite constellation systems and integration of SATCOM architecture in 5G terrestrial scenarios, NOMA and its variants, e.g., power domain, code domain (sparse code multiple access (SCMA), low density spreading (LDS), multi-user shared access (MUSA)), pattern division, lattice partition and bit division [196], can be extended to SATCOM systems for attaining high throughput with the available bandwidth. The application of NOMA in layered division multiplexing, a new multiplexing scheme proposed by the advanced television systems committee (ATSC) for terrestrial digital TV broadcasting systems, is a good example [204]. Similarly, for scheduling of user terminals, suboptimal algorithms have been proposed to maximize the overloaded SATCOM system capacity [205]. In [206], the overlay coding technique is used to mitigate interference in multi-beam SATCOM systems based on NOMA channel access techniques. In [207], the channel conditions for different users have been studied for relaying data in a satellite-terrestrial relay system. A user with a high channel gain can relay data to another

user with poor channel condition resulting in low outage probability and greater ergodic capacity compared to OMA. In recent advancements, integration of NOMA and coordinated multi-point (CoMP) transmissions has been studied to measure the performance of NOMA in terms of ergodic rates, outage probabilities, maximizing sum-rates, and dynamic power allocation strategies [208]–[210]. In this regard two popular schemes have been analyzed, the Alamouti-NOMA and the joint-transmission NOMA [211]. Recent advancements in satellite-aerial-terrestrial networks accommodate unmanned aerial vehicle (UAV) enabled networks to enhance system throughput, maximize the terrestrial user's achievable rate, to relay the desired information, and interference cancellation [22], [212]–[214]. However, extension of NOMA for terrestrial 5G networks to SATCOM systems and constellations may require modification to the channel propagation models due to similar path loss for multiple ground stations [3], user pairing strategies based on channel gains [82], complexity of design with respect to multi-beam capability, design constraints (on-board power resources, propagation delays, signal distortion and mobility issues), optimal beam forming and capability in terms of allocation of onboard resources [215].

The demand for higher capacity, low latency, ultra-high reliability and massive connectivity for next generation wireless communication networks requires next generation multiple access schemes [216]. In the unsourced multiple access scheme [217], a sparse Tanner graph based transmission scheme has been proposed for MIMO channels. The proposed scheme has a lower complexity and outperforms the compressed sensing based unsourced multiple access scheme. The device selection and resource allocation for non-uniform device based on federated meta-learning algorithm outperforms baseline algorithms [218]. Considering the MAC, the QMIX-advanced listen-before-talk (QLBT) protocol achieves improved throughput, smaller delay and low jitter [219]. The proposed protocol uses multi-agent reinforcement learning for centralized training and decentralized execution. Rainbow link, a novel FDMA-based access scheme, maps resources to specific spatial directions, creating subsets of OFDMA subcarriers and appreciate beamforming gain for the complete array [220]. The proposed scheme does not require explicit beam training, and supports random access. Thus, is promising for latency critical applications. The RSMA technique for visible light communication compensates for the limited modulation bandwidth for single and multi-cell networks and has proven to be energy efficient compared to benchmark schemes [221]. To summarize, next-generation multiple access (NGMA) requires extensive research efforts in the field of emerging multiplexing schemes such as NOMA and SDMA, reconfigurable and advanced modulation and channel coding schemes, and application of machine learning tools for NGMA requirements.

Optical technologies offer an alternative approach to overcome RF band bottleneck in SATCOM by providing 10 THz bandwidth and have the potential to be the enabling technology for future digitization revolution. However, atmospheric attenuation in the RF band at higher frequencies is the main challenge which can be mitigated using the site di-

versity approach. The prospect of optical technologies in high throughput satellite systems can deliver high data rates (Tbps throughput) for GEO communication systems using dense wave division multiplexing (DWDM) [222]. For homodyne receivers and using QPSK modulation, 80 Gbps throughput has been achieved. The emerging needs of global coverage and throughput cannot be met using existing GEO satellite systems due to the limited RF band spectrum, i.e., 1 GHz bandwidth in the Ku -band (12-18 GHz) and 6 GHz in the Ka -band (27-40 GHz) which is shared by government, military and terrestrial fixed services. Higher bands, e.g., Q/V (40-50 GHz) allocate 8 GHz bandwidth for broadband services without exclusive allocation and also share the spectrum with military and terrestrial services [223].

SATCOM is based on the pointing mechanism. Different point-ahead angles (PAA) exist for GEO (18 μ rad), MEO (20-30 μ rad) and LEO (30-60 μ rad) satellites. A large PAA limits the pointing performance and imposes higher transmission power to overcome the divergence of the uplink beam. A link budget performance for GEO, LEO and MEO satellites depicts that 2.3 dB less transmitted power is required for LEO satellites compared to GEO satellites. For MEO satellites, the difference is 6–14 dB depending on the pointing mechanism, i.e., DL-tracking pointing and open-loop pointing. The main challenge for optical communication is atmospheric turbulence. To demonstrate the terabit scale between the GEO satellite and the optical ground station, a 2 degree elevation was considered in [224] to emulate the optical SATCOM link under the worst turbulence conditions. Forty DWDM channel with 100 GHz capacity each, modulated with On-Off keying (OOK) at modulation rate of 43 Gbps achieved a throughput of 1.72 Tbps. A pre-amplification of the coherent reception increased throughput to 13.16 Tbps using the DP-16 QAM modulation scheme for 53 channels modulated at 34 GBaud. Moreover, digital homodyne processing used for an extra channel of 100 GHz with 80 Gbps and QPSK modulation scheme. Multiple modulation schemes i.e., DP-QPSK, DP-8QAM and DP-16QAM were tested to determine successful use of higher order modulation schemes under strong atmospheric turbulence conditions.

In integrated SATCOM systems, and 5G and 6G networks, terrestrial networks are an attractive option to resolve satellite spectrum scarcity and provide high speed, seamless global services. The integration of satellites with non-terrestrial platforms (e.g., UAVs), high altitude platforms, and terrestrial platforms can enhance the wireless coverage, provide high-speed data rates with broadband connectivity, and act as a backup route for sustainable communication [25], [29], [31], [197], [225], [226]. However, efficient network access schemes are required that can overcome interference and congestion issues. In this regard, satellite spectrum sensing capability, grouping of users into primary and secondary, satellite beamforming technology, and satellite receiver design are some areas that need to be explored further.

B. Modulation Schemes

Digital modulation schemes with constant envelope (M -PSK and M -FSK) are suitable for SATCOM systems since

these are not severely distorted in the non-linear region of the high power amplifiers. For $M = 2^k$ with $k \in \{1, 2, \dots, 6\}$, M -PSK has shown good spectral efficiency. But with a higher modulation index, the signals are more sensitive to channel impairments due to the proximity of the constellation points. Moreover, the system complexity is also increased. The most popular schemes used for SATCOM are QPSK and its variant $\pi/4$ -QPSK and OQPSK due to their better spectral efficiency and resilience against channel noise and interference. FSK schemes are not suitable for SATCOM systems due to the low bandwidth efficiency. MSK and GFSK are two popular FSK schemes used in SATCOM systems where bandwidth efficiency is traded-off for lower system complexity. A higher M depicts better data transmission capacity, but for the allocated bandwidth, the receiver shows high sensitivity due to the reduced frequency deviation. It is for this reason that a continuous phase modulation scheme is preferred to avoid abrupt transitions in the phase which result in the lower side-lobe energy.

For non-constant envelopes, e.g., M -QAM, the information is decoded by varying both the amplitude and phase. As compared to 16PSK, 16-QAM depicts higher resilience to channel impairments due to the greater separation between the constellation points. However, non-constant envelope requires wider linear operational range for high power amplifier (HPA) resulting in increased cost and complexity. M -APSK for $M = 16, 32, 64$ is also used in modern SATCOM systems. APSK can overcome signal impairment issues suffered by PSK and requires a smaller linear operational range for HPA compared to QAM due to fewer amplitude levels. For example, 16-APSK has only 2 amplitude levels while QAM has 3. Similarly, 32-APSK has 3 amplitude levels compared to 5 levels of QAM. However, these schemes require high-end algorithms for packet detection and synchronization. Orthogonal time-frequency space (OTFS), a recent two-dimensional modulation scheme, can provide high speed, high frequency and large antenna array transmissions [227]. OTFS can overcome the time-varying issues of the channel and can provide near-constant channel gain due to time-independent characteristics [196].

Adaptive coding and modulation (ACM) for enhancing transmission efficiency and error tolerance is a modern technique. It improves the system power efficiency, reduces inter-user interferences and can meet variable data rate requirements [27], [228]. ACM can provide higher data rates, particularly for small satellites which operate in the NGSO where channel SNR is time-varying. Selective retransmission under ACM coupled with OFDM signals can enhance the spectral efficiency by partial retransmission of failed packets [228]. ISL will play a pivotal role for advanced SATCOM systems. Enhanced spectral efficiency, power efficiency and improved BER can be achieved using ACM for these links. ACM can achieve 6 $\times$ the throughput of constant modulation and coding scheme [229]. To overcome the distortions in ACM due to QAM, higher-order partial response continuous phase modulation can be used to enhance power and spectral efficiency [230]. The higher-frequency bands with ACM may be used [126]. The authors have stated that apart from a few

theoretical studies, there have been no in-depth analyses of ACM in terms of optimization of the adaptive modulation and coding technology. Moreover, the scarcity problem for RF band and synchronization issue may be reduced by using visible light communication, having higher data rates with laser communication.

C. Channel Encoding Schemes

Reliable and low latency communication are crucial for future communication systems involving satellite constellations, 5G, and 6G, which highlights the importance of channel coding. While TB-CC and TC have been popular choices for LTE, LDPC codes, PC, TB-CC, and TC are recommended for 5G URLLC and eMBB systems. Recently, rateless codes have emerged as a potential contender for future wireless communication systems [32]. LDPC codes and TC have shown exceptionally good performance for block lengths $> 2,000$ bits [125]–[133]. But these codes lead to high latency. The design for short block-lengths is an open problem due to conflicting requirements of reliability and low latency [19], [101], [133]–[139]. Latency in channel coding can be reduced by optimizing the contributing factors such as time-transmit latency, processing latency and retransmission time [137] that depend on throughput and error performance of the channel decoder.

QC-LDPC codes are a popular choice for the analysis of the short block-length regime [152] which outperform TC with a lower complexity [231], [232]. A recent variant of QC-LDPC codes, the so-called protograph-based raptor-like LDPC (PBRL-LDPC) codes is a promising choice for low latency applications due to fewer decoding iterations required [233]. The stringent requirements of URLLC and mMTC can be met by PC which have depicted reliability of 99.99% with very low power requirements compared to LDPC, TC and TB-CC codes [234]. Parity-check PC is the strongest candidate from variants of PC including extended BCH (eBCH) and parity-check-concatenated PC for all block lengths and code rates [235]. The concatenation of PC, LDPC, and spinal codes (variant of rateless codes), and hybridization of PC and turbo product codes have depicted promising results for low latency, reduced implementation complexity, and enhanced power efficiency [236]–[238]. CRC-aided spinal codes with TB structures reduce decoding complexity compared to conventional spinal codes and therefore need further exploration [239].

Extensive research on short-block length codes has revealed that Polyansky-Poor-Verdu (PPV) limits have not been achieved by any channel codes [240], [241]. Hybridization of PC and LDPC codes needs to be explored further to meet the requirements of URLLC and mMTC services [235]. The complex implementation of TC limits their application for future wireless communication systems. However, if their complexity can be reduced, they may prove useful for high throughput wireless communication systems [242], [243].

D. SDR Implementation of SATCOM Systems

Future wireless communication systems, i.e., integrated satellite systems, 5G and 6G require reconfigurability, adaptivity and flexibility. For advanced GSO satellite systems, SDRs

can be useful for optimization of the available bandwidth and modulation schemes, multi-satellite transmissions, high throughput using multi-beam transmissions, and interference mitigation capabilities [172], [182]–[185]. In addition to interference mitigation, SDR testbeds can also be used for airplane connectivity [180], [181]. Pre-coding techniques for power minimization and multi-user multi-communication system can be analysed using SDR testbeds for advanced SATCOM systems [244], [245].

The evolution in payload, telecommunication architecture, communication, and network protocols, all indicate to an increased use of small satellites for telecommunication services [126]. In multi-constellation systems, delay-tolerant networking (DTN) protocols can use ring road architecture. However, non-DTN and DTN architectures demand proper integration for successful communication between multiple space and ground segments. The softwarisation of the network can be achieved via network function virtualization (NFV) and software defined network (SDN) architectures. Moreover, NFV/SDN protocols may play a key role in 5G communication, if fully investigated.

Advances in satellite constellation systems have increased the coverage of small satellite systems. SDRs provide the platform to explore aspects of practical implementation [102]. FPGA compatible SDR payloads provide reconfigurable platforms for emerging CubSAT platforms (GomSpace, NanoAvionics, AlenSpace, ISISpace). LEO satellites offer smaller propagation delay, lower atmospheric attenuation, coverage area limitations, free space loss and lower launching cost. SDRs can:

- be used to explore the viability of higher frequency bands, laser communication, and novel communication protocols and architecture for future systems [30], [114], [126], [194];
- enhance throughput and spectral efficiency by using multi-beam and multi-band heterogeneous satellite transmissions for multi-orbit satellite systems [28];
- be used for analysing the Doppler shift in inter-satellite communication systems and satellite-terrestrial advanced networking systems;
- assist in modelling the multi-constellation receivers capable of supporting multiple modulation schemes [190];
- be used to explore modern channel encoding techniques, waveform designs for fulfilling the stringent requirements of URLLC, mMTC and eMBB;
- be used to explore practical implementation of LDPC codes for advanced satellite systems [174], [175];
- provide the capability of pulse shaping optimization and multi-component signalling for designing modern waveforms [178];
- be used to study the interference mitigation capabilities and techniques for SATCOM systems [179];
- be utilized with heterogeneous processors and multi-core processors to evaluate interference mitigation techniques and to enhance the computational performance [162], [163], [171], [172].

VII. CONCLUSIONS

In this paper, we have discussed advancements in SATCOM systems, in particular, with a focus on requirements of 5G and 6G communication systems. We have discussed the enabling technologies and advancements in communication protocols, their benefits, limitations, and open problems with respect to multiplexing, modulations, channel encoding and SDR-based platforms for SATCOM systems. We have underscored that the rapidly growing demand for various services, and the conflicting requirement of higher throughput, lower latency and massive device connections require integration of space and terrestrial services at multi-layer level. Moreover, the rapid advancements in satellite constellations, inter-satellite and intra-satellite communication require integration of existing and next generation architecture to provide services and meet quality of service standards. We have highlighted the extensive research in adaptive modulation and coding schemes, URLLC channel encoding techniques for real time applications, LDPC and PC, modern multiplexing schemes such as OFDMA, and NOMA for enhancing spectral and power efficiency and the need for inter-satellite communication systems to meet the growing demand for network services and applications.

REFERENCES

- [1] Y. Vasavada, R. Gopal, C. Ravishankar, G. Zakaria, and N. BenAmmar, "Architectures for next generation high throughput satellite systems," *International Journal of Satellite Communications and Networking*, vol. 34, no. 4, pp. 523–546, Jan 2016.
- [2] Y. Hu and V. Li, "Satellite-based internet: a tutorial," *IEEE Communications Magazine*, vol. 39, no. 3, pp. 154–162, Mar 2001.
- [3] A. I. Perez-Neira, M. A. Vazquez, M. B. Shankar, S. Maleki, and S. Chatzinotas, "Signal processing for high-throughput satellites: Challenges in new interference-limited scenarios," *IEEE Signal Processing Magazine*, vol. 36, no. 4, pp. 112–131, Jul 2019.
- [4] N. Mazzali, M. R. B. Shankar, and B. Ottersten, "On-board signal predistortion for digital transparent satellites," in *2015 IEEE 16th International Workshop on Signal Processing Advances in Wireless Communications (SPAWC)*, Jun 2015.
- [5] C. Politis, S. Maleki, C. Tsinos, S. Chatzinotas, and B. Ottersten, "Weak interference detection with signal cancellation in satellite communications," in *2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Mar 2017.
- [6] G. Araniti, I. Bisio, M. D. Sanctis, A. Orsino, and J. Cosmas, "Multimedia content delivery for emerging 5G-satellite networks," *IEEE Transactions on Broadcasting*, vol. 62, no. 1, pp. 10–23, Mar 2016.
- [7] O. Erceetin, M. O. Ball, and L. Tassiulas, "Next generation satellite systems for aeronautical communications," *International Journal of Satellite Communications and Networking*, vol. 22, no. 2, pp. 157–179, Feb 2004.
- [8] F. Alagoz, O. Korcak, and A. Jamalipour, "Exploring the routing strategies in next-generation satellite networks," *IEEE Wireless Communications*, vol. 14, no. 3, pp. 79–88, Jun 2007.
- [9] J. Bao, B. Zhao, W. Yu, Z. Feng, C. Wu, and Z. Gong, "OpenSAN," *ACM SIGCOMM Computer Communication Review*, vol. 44, no. 4, pp. 347–348, Feb 2015.
- [10] J. Mukherjee and B. Ramamurthy, "Communication technologies and architectures for space network and interplanetary internet," *IEEE Communications Surveys & Tutorials*, vol. 15, no. 2, pp. 881–897, 2013.
- [11] R. Ferrús, H. Koumaras, O. Sallent, G. Agapiou, T. Rasheed, M.-A. Kourtis, C. Boustie, P. Gélard, and T. Ahmed, "SDN/NFV-enabled satellite communications networks: Opportunities, scenarios and challenges," *Physical Communication*, vol. 18, pp. 95–112, Mar 2016.
- [12] C. Niephaus, M. Kretschmer, and G. Ghinea, "QoS provisioning in converged satellite and terrestrial networks: A survey of the state-of-the-art," *IEEE Communications Surveys & Tutorials*, vol. 18, no. 4, pp. 2415–2441, 2016.
- [13] D. Kreutz, F. M. V. Ramos, P. E. Verissimo, C. E. Rothenberg, S. Azodolmolky, and S. Uhlig, "Software-defined networking: A comprehensive survey," *Proceedings of the IEEE*, vol. 103, no. 1, pp. 14–76, Jan 2015.
- [14] N. Chowdhury and R. Boutaba, "Network virtualization: state of the art and research challenges," *IEEE Communications Magazine*, vol. 47, no. 7, pp. 20–26, Jul 2009.
- [15] R. Mijumbi, J. Serrat, J.-L. Gorricho, N. Bouten, F. D. Turck, and R. Boutaba, "Network function virtualization: State-of-the-art and research challenges," *IEEE Communications Surveys & Tutorials*, vol. 18, no. 1, pp. 236–262, 2016.
- [16] G. Baldini, T. Sturman, A. R. Biswas, R. Leschhorn, G. Godor, and M. Street, "Security aspects in software defined radio and cognitive radio networks: A survey and a way ahead," *IEEE Communications Surveys & Tutorials*, vol. 14, no. 2, pp. 355–379, 2012.
- [17] S. K. Sharma, T. E. Bogale, L. B. Le, S. Chatzinotas, X. Wang, and B. Ottersten, "Dynamic spectrum sharing in 5G wireless networks with full-duplex technology: Recent advances and research challenges," *IEEE Communications Surveys & Tutorials*, vol. 20, no. 1, pp. 674–707, 2018.
- [18] A. F. Demir, M. Elkourdi, M. Ibrahim, and H. Arslan, "Waveform design for 5G and beyond," *arXiv preprint arXiv:1902.05999*, 2019.
- [19] S. Cheng, "Comparative study on 5g communication channel coding technology," in *3rd International Conference on Mechatronics Engineering and Information Technology (ICMEIT 2019)*, *Advances in Computer Science Research*, vol. 87, 2019, pp. 74–77.
- [20] P. K. Sangdeh and H. Zeng, "Overview of multiplexing techniques in wireless networks," in *Multiplexing*. IntechOpen London, 2019.
- [21] T. Ferrer, S. Céspedes, and A. Becerra, "Review and evaluation of MAC protocols for satellite IoT systems using nanosatellites," *Sensors*, vol. 19, no. 8, p. 1947, Apr 2019.
- [22] M. Hua, Y. Wang, M. Lin, C. Li, Y. Huang, and L. Yang, "Joint CoMP transmission for UAV-aided cognitive satellite terrestrial networks," *IEEE Access*, vol. 7, pp. 14 959–14 968, 2019.
- [23] S. Chen, Y.-C. Liang, S. Sun, S. Kang, W. Cheng, and M. Peng, "Vision, requirements, and technology trend of 6G: How to tackle the challenges of system coverage, capacity, user data-rate and movement speed," *IEEE Wireless Communications*, vol. 27, no. 2, pp. 218–228, Apr 2020.
- [24] F. Conceição, M. Gomes, V. Silva, R. Dinis, A. Silva, and D. Castanheira, "A survey of candidate waveforms for beyond 5G systems," *Electronics*, vol. 10, no. 1, p. 21, Dec 2020.
- [25] A. Vanelli-Coralli, A. Guidotti, T. Foggi, G. Colavolpe, and G. Montorsi, "5G and beyond 5G non-terrestrial networks: trends and research challenges," in *2020 IEEE 3rd 5G World Forum (5GWF)*, Sep 2020.
- [26] A. Guidotti, S. Cioni, G. Colavolpe, M. Conti, T. Foggi, A. Mengali, G. Montorsi, A. Piemontese, and A. Vanelli-Coralli, "Architectures, standardisation, and procedures for 5G satellite communications: A survey," *Computer Networks*, vol. 183, p. 107588, Dec 2020.
- [27] Y. Wang, W. Liu, and L. Fang, "Adaptive modulation and coding technology in 5G system," in *2020 International Wireless Communications and Mobile Computing (IWCMC)*, Jun 2020.
- [28] F. Li, K.-Y. Lam, H.-H. Chen, and N. Zhao, "Spectral efficiency enhancement in satellite mobile communications: A game-theoretical approach," *IEEE Wireless Communications*, vol. 27, no. 1, pp. 200–205, Feb 2020.
- [29] J.-H. Lee, J. Park, M. Bennis, and Y.-C. Ko, "Integrating LEO satellite and UAV relaying via reinforcement learning for non-terrestrial networks," in *GLOBECOM 2020-2020 IEEE Global Communications Conference*, 2020, pp. 1–6.
- [30] O. Kodheli, E. Lagunas, N. Maturo, S. K. Sharma, B. Shankar, J. F. M. Montoya, J. C. M. Duncan, D. Spano, S. Chatzinotas, S. Kisseleff, J. Querol, L. Lei, T. X. Vu, and G. Goussetis, "Satellite communications in the new space era: A survey and future challenges," *IEEE Communications Surveys & Tutorials*, vol. 23, no. 1, pp. 70–109, 2021.
- [31] B. Evans, N. Wang, Y. Rahulan, S. Kumar, J. Cahill, M. Kavanagh, S. Watts, D.-K. Chau, Y. Begassat, A.-P. Brunel, T. Masson, and M. Diarrah, "An integrated satellite-terrestrial 5G network and its use to demonstrate 5G use cases," *International Journal of Satellite Communications and Networking*, vol. 39, no. 4, pp. 358–379, Feb 2021.
- [32] M. Indoonundon and T. P. Fowdur, "Overview of the challenges and solutions for 5G channel coding schemes," *Journal of Information and Telecommunication*, vol. 5, no. 4, pp. 460–483, Jul 2021.
- [33] P. J. Honnaiah, N. Maturo, S. Chatzinotas, S. Kisseleff, and J. Krause, "Demand-based adaptive multi-beam pattern and footprint planning for

- high throughput GEO satellite systems,” *IEEE Open Journal of the Communications Society*, vol. 2, pp. 1526–1540, 2021.
- [34] I. Oppermann and B. S. Vucetic, “Coded direct sequence spread spectrum systems over satellite channels using an adaptive receiver,” in *Proceedings of ISSSTA’95 International Symposium on Spread Spectrum Techniques and Applications*, vol. 3, Sep 1996, pp. 940–944.
 - [35] G. Heinrichs, G. Rohmer, and F. Henkel, “A fully digital direct-sequence spread-spectrum receiver architecture for GNSS applications,” in *2000 IEEE Sixth International Symposium on Spread Spectrum Techniques and Applications*, vol. 2, 2000, pp. 510–514.
 - [36] M. Pursley, “Direct-sequence spread-spectrum communications for multipath channels,” *IEEE Transactions on Microwave Theory and Techniques*, vol. 50, no. 3, pp. 653–661, Mar 2002.
 - [37] Y. S. Chan, J. W. Modestino, Q. Qu, and X. Fan, “An end-to-end embedded approach for multicast/broadcast of scalable video over multiuser CDMA wireless networks,” *IEEE Transactions on Multimedia*, vol. 9, no. 3, pp. 655–667, Apr 2007.
 - [38] Z. Wei and R. Tao, “A residual interference mitigation technique for frequency-domain interference suppression in dsss communication systems,” in *2017 9th International Conference on Wireless Communications and Signal Processing (WCSP)*, Oct 2017, pp. 1–6.
 - [39] Z. Yao, F. Guo, J. Ma, and M. Lu, “Orthogonality-based generalized multicarrier constant envelope multiplexing for DSSS signals,” *IEEE Transactions on Aerospace and Electronic Systems*, vol. 53, no. 4, pp. 1685–1698, Aug 2017.
 - [40] J. Khalife, K. Shamaei, and Z. M. Kassas, “Navigation with cellular CDMA signals—part I: Signal modeling and software-defined receiver design,” *IEEE Transactions on Signal Processing*, vol. 66, no. 8, pp. 2191–2203, Apr 2018.
 - [41] Z. Zhang, F. Tian, F. Zeng, S. He, Y. Liu, and G. Xuan, “Performance comparison of two spread-spectrum-based wireless video transmission schemes,” in *2016 9th International Congress on Image and Signal Processing, BioMedical Engineering and Informatics (CISP-BMEI)*, Oct 2016, pp. 1216–1220.
 - [42] F. Xiong, “Modem techniques in satellite communications,” *IEEE Communications Magazine*, vol. 32, no. 8, pp. 84–98, Aug 1994.
 - [43] A. J. Viterbi, “Very low rate convolution codes for maximum theoretical performance of spread-spectrum multiple-access channels,” *IEEE Journal on Selected Areas in Communications*, vol. 8, no. 4, pp. 641–649, May 1990.
 - [44] T. Ikegami and M. B. Pursley, “The effects of transmitter filtering on the performance of direct sequence spread spectrum multiple access systems,” in *Proceedings of IEEE 3rd International Symposium on Spread Spectrum Techniques and Applications (ISSSTA’94)*, Jul 1994, pp. 569–573.
 - [45] P. Bergadà, R. M. Alsina-Pagès, J. L. Pijoan, M. Salvador, J. R. Regué, D. Badia, and S. Graells, “Digital transmission techniques for a long haul HF link: DSSS versus OFDM,” *Radio Science*, vol. 49, no. 7, pp. 518–530, Jul 2014.
 - [46] R. M. Alsina-Pagès, D. Badia, P. Bergadà, M. Salvador, J. L. Pijoan, and M. Hervás, “Spread spectrum high performance techniques for a long haul high frequency link,” *IET Communications*, vol. 9, no. 8, pp. 1048–1053, May 2015.
 - [47] L. K. Nguyen, R. B. Wells, D. H. N. Nguyen, and N. H. Tran, “Outage probability analysis of the protected tactical waveform (PTW) on the return link,” in *MILCOM 2017 - 2017 IEEE Military Communications Conference (MILCOM)*, Oct 2017.
 - [48] R. De Gaudenzi and O. del Rio Herrero, “Advances in random access protocols for satellite networks,” in *2009 International Workshop on Satellite and Space Communications*, 2009, pp. 331–336.
 - [49] C. Kissling and A. M. Dlr, “On the integration of random access and DAMA channels for the return link of satellite networks,” in *2013 IEEE International Conference on Communications (ICC)*, 2013, pp. 4282–4287.
 - [50] Y.-P. E. Wang, X. Lin, A. Adhikary, A. Grovlen, Y. Sui, Y. Blankenship, J. Bergman, and H. S. Razaghi, “A primer on 3GPP narrowband internet of things,” *IEEE Communications Magazine*, vol. 55, no. 3, pp. 117–123, Mar 2017.
 - [51] L. Vangelista, A. Zanella, and M. Zorzi, “Long-range IoT technologies: The dawn of LoRa™,” in *Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering*, Springer International Publishing, 2015, pp. 51–58.
 - [52] S. Cluzel, M. Dervin, J. Radzik, S. Cazalens, C. Baudoin, and D. Dragomirescu, “Physical layer abstraction for performance evaluation of LEO satellite systems for IoT using time-frequency ALOHA scheme,” in *2018 IEEE Global Conference on Signal and Information Processing (GlobalSIP)*, 2018, pp. 1076–1080.
 - [53] M. Krondorf, M. Goblirsch, R. de Gaudenzi, G. Cocco, N. Toptsidis, and G. Acar, “Towards the implementation of advanced random access schemes for satellite IoT,” *International Journal of Satellite Communications and Networking*, vol. 38, no. 2, pp. 177–199, Nov 2019.
 - [54] R. D. Gaudenzi, O. D. R. Herrero, G. Gallinaro, S. Cioni, and P.-D. Arapoglou, “Random access schemes for satellite networks, from VSAT to M2M: a survey,” *International Journal of Satellite Communications and Networking*, vol. 36, no. 1, pp. 66–107, Dec 2016.
 - [55] A. Mengali, R. D. Gaudenzi, and C. Stefanovic, “On the modeling and performance assessment of random access with SIC,” *IEEE Journal on Selected Areas in Communications*, vol. 36, no. 2, pp. 292–303, Feb 2018.
 - [56] R. D. Gaudenzi, O. del Rio Herrero, G. Acar, and E. G. Barrabes, “Asynchronous contention resolution diversity ALOHA: Making CRDSA truly asynchronous,” *IEEE Transactions on Wireless Communications*, vol. 13, no. 11, pp. 6193–6206, Nov 2014.
 - [57] E. Casini, R. D. Gaudenzi, and O. Herrero, “Contention resolution diversity slotted ALOHA (CRDSA): An enhanced random access scheme for satellite access packet networks,” *IEEE Transactions on Wireless Communications*, vol. 6, no. 4, pp. 1408–1419, Apr 2007.
 - [58] T.-D. Chiueh and P.-Y. Tsai, *OFDM baseband receiver design for wireless communications*. John Wiley & Sons, 2008.
 - [59] C. Balint and G. Budura, “OFDM-based multi-carrier waveforms performances in 5G,” in *2018 International Symposium on Electronics and Telecommunications (ISETC)*, 2018, pp. 1–4.
 - [60] B. Somasekhar, C. S. Kishore, D. T. R. Sai, C. Asha, S. Manasa, and C. Nalanda, “Inter carrier interference mitigation in space frequency block code MIMO-OFDM system,” *International Journal of Advanced Science and Technology*, vol. 23, no. 1, pp. 70–109, 2020.
 - [61] S. Gökceli, H. Alakoca, S. T. Başaran, and G. K. Kurt, “OFDMA-based network-coded cooperation: design and implementation using software-defined radio nodes,” *EURASIP Journal on Advances in Signal Processing*, vol. 2016, no. 1, Jan 2016.
 - [62] X. Zhang, M. Jia, L. Chen, J. Ma, and J. Qiu, “Filtered-OFDM-enabler for flexible waveform in the 5th generation cellular networks,” in *2015 IEEE Global Communications Conference (GLOBECOM)*, 2015, pp. 1–6.
 - [63] X. Zhang, L. Chen, J. Qiu, and J. Abdoli, “On the waveform for 5G,” *IEEE Communications Magazine*, vol. 54, no. 11, pp. 74–80, Nov 2016.
 - [64] H. Chen, J. Hua, F. Li, F. Chen, and D. Wang, “Interference analysis in the asynchronous f-OFDM systems,” *IEEE Transactions on Communications*, vol. 67, no. 5, pp. 3580–3596, May 2019.
 - [65] N. Michailow, I. Gaspar, S. Krone, M. Lentmaier, and G. Fettweis, “Generalized frequency division multiplexing: Analysis of an alternative multi-carrier technique for next generation cellular systems,” in *2012 International Symposium on Wireless Communication Systems (ISWCS)*, Aug 2012.
 - [66] B. M. Alves, L. L. Mendes, D. A. Guimaraes, and I. S. Gaspar, “Performance of GFDM over frequency-selective channels,” in *Proc. Int. Workshop Telecommun.*, 2013, pp. 1–7.
 - [67] N. Michailow, M. Matthe, I. S. Gaspar, A. N. Caldeilla, L. L. Mendes, A. Festag, and G. Fettweis, “Generalized frequency division multiplexing for 5th generation cellular networks,” *IEEE Transactions on Communications*, vol. 62, no. 9, pp. 3045–3061, Sep 2014.
 - [68] R. Datta, N. Michailow, M. Lentmaier, and G. Fettweis, “GFDM interference cancellation for flexible cognitive radio PHY design,” in *2012 IEEE Vehicular Technology Conference*, 2012, pp. 1–5.
 - [69] M. Series, “IMT vision—framework and overall objectives of the future development of IMT for 2020 and beyond,” *Recommendation ITU*, vol. 2083, p. 21, 2015.
 - [70] W. D. Dias, L. L. Mendes, and J. J. P. C. Rodrigues, “Low complexity GFDM receiver for frequency-selective channels,” *IEEE Communications Letters*, vol. 23, no. 7, pp. 1166–1169, Jul 2019.
 - [71] B. Lim and Y.-C. Ko, “Multiuser interference cancellation for GFDM with timing and frequency offsets,” *IEEE Transactions on Communications*, vol. 67, no. 6, pp. 4337–4349, Jun 2019.
 - [72] A. Mohammadian, C. Tellambura, and M. Valkama, “Analysis of self-interference cancellation under phase noise, CFO, and IQ imbalance in GFDM full-duplex transceivers,” *IEEE Transactions on Vehicular Technology*, vol. 69, no. 1, pp. 700–713, Jan 2020.
 - [73] I. Gaspar, N. Michailow, A. Navarro, E. Ohlmer, S. Krone, and G. Fettweis, “Low complexity GFDM receiver based on sparse frequency domain processing,” in *2013 IEEE 77th Vehicular Technology Conference*, Jun 2013.
 - [74] J. Nunes, P. Bento, M. Gomes, R. Dinis, and V. Silva, “Block-windowed burst OFDM: a high-efficiency multicarrier technique,” *Electronics Letters*, vol. 50, no. 23, pp. 1757–1759, Nov 2014.

- [75] T. Fernandes, M. Gomes, V. Silva, and R. Dinis, "Time-interleaved block-windowed burst OFDM," in *2016 IEEE 84th Vehicular Technology Conference*, Sep 2016.
- [76] T. Fernandes, A. Pereira, M. Gomes, V. Silva, and R. Dinis, "A new hybrid multicarrier transmission technique with iterative frequency domain detection," *Physical Communication*, vol. 27, pp. 7–16, Apr 2018.
- [77] F. Conceição, M. Gomes, V. Silva, and R. Dinis, "Time overlapping TIBWB-OFDM symbols for peak-to-average power ratio reduction," in *Proceedings of the 11th Conference on Telecommunications (ConfTele 2019)*, Lisbon, Portugal, 2019, pp. 26–28.
- [78] Y. Liang, J. Ren, and T. Li, "Secure OFDM system design and capacity analysis under disguised jamming," *IEEE Transactions on Information Forensics and Security*, vol. 15, pp. 738–752, 2020.
- [79] M. R. Hassan, G. C. Karmakar, J. Kamruzzaman, and B. Srinivasan, "Exclusive use spectrum access trading models in cognitive radio networks: A survey," *IEEE Communications Surveys & Tutorials*, vol. 19, no. 4, pp. 2192–2231, 2017.
- [80] S. K. Sharma, T. E. Bogale, S. Chatzinotas, B. Ottersten, L. B. Le, and X. Wang, "Cognitive radio techniques under practical imperfections: A survey," *IEEE Communications Surveys & Tutorials*, vol. 17, no. 4, pp. 1858–1884, 2015.
- [81] C. Yang, J. Li, M. Guizani, A. Anpalagan, and M. El-kashlan, "Advanced spectrum sharing in 5G cognitive heterogeneous networks," *IEEE Wireless Communications*, vol. 23, no. 2, pp. 94–101, Apr 2016.
- [82] S. R. Islam, N. Avazov, O. A. Dobre, and K.-S. Kwak, "Power-domain non-orthogonal multiple access (noma) in 5G systems: Potentials and challenges," *IEEE Communications Surveys & Tutorials*, vol. 19, no. 2, pp. 721–742, 2016.
- [83] L. Lei, D. Yuan, C. K. Ho, and S. Sun, "Power and channel allocation for non-orthogonal multiple access in 5G systems: Tractability and computation," *IEEE Transactions on Wireless Communications*, vol. 15, no. 12, pp. 8580–8594, Dec 2016.
- [84] L. Lei, D. Yuan, and P. Varbrand, "On power minimization for non-orthogonal multiple access (NOMA)," *IEEE Communications Letters*, vol. 20, no. 12, pp. 2458–2461, Dec 2016.
- [85] L. Lei, L. You, Y. Yang, D. Yuan, S. Chatzinotas, and B. Ottersten, "Load coupling and energy optimization in multi-cell and multi-carrier NOMA networks," *IEEE Transactions on Vehicular Technology*, vol. 68, no. 11, pp. 11 323–11 337, Nov 2019.
- [86] X. Zhu, C. Jiang, L. Kuang, N. Ge, and J. Lu, "Non-orthogonal multiple access based integrated terrestrial-satellite networks," *IEEE Journal on Selected Areas in Communications*, vol. 35, no. 10, pp. 2253–2267, Oct 2017.
- [87] Z. Lin, M. Lin, J.-B. Wang, T. de Cola, and J. Wang, "Joint beamforming and power allocation for satellite-terrestrial integrated networks with non-orthogonal multiple access," *IEEE Journal of Selected Topics in Signal Processing*, vol. 13, no. 3, pp. 657–670, Jun 2019.
- [88] X. Liu, K.-Y. Lam, F. Li, J. Zhao, L. Wang, and T. S. Durrani, "Spectrum sharing for 6G integrated satellite-terrestrial communication networks based on NOMA and CR," *IEEE Network*, vol. 35, no. 4, pp. 28–34, Jul 2021.
- [89] X. Zhang, B. Zhang, K. An, G. Wu, Y. Jia, S. Qi, and D. Guo, "NOMA-based proactive content caching in hybrid satellite-aerial-terrestrial networks," in *2021 IEEE Wireless Communications and Networking Conference Workshops (WCNCW)*, 2021, pp. 1–6.
- [90] R. Ge, D. Bian, J. Cheng, K. An, J. Hu, and G. Li, "Joint user pairing and power allocation for NOMA-based GEO and LEO satellite network," *IEEE Access*, vol. 9, pp. 93 255–93 266, 2021.
- [91] M. Schubert and H. Boche, "Solution of the multiuser downlink beamforming problem with individual SINR constraints," *IEEE Transactions on Vehicular Technology*, vol. 53, no. 1, pp. 18–28, Jan 2004.
- [92] A. Gershman, N. Sidiropoulos, S. Shahbazpanahi, M. Bengtsson, and B. Ottersten, "Convex optimization-based beamforming," *IEEE Signal Processing Magazine*, vol. 27, no. 3, pp. 62–75, May 2010.
- [93] Y.-F. Liu, Y.-H. Dai, and Z.-Q. Luo, "Coordinated beamforming for MISO interference channel: Complexity analysis and efficient algorithms," *IEEE Transactions on Signal Processing*, vol. 59, no. 3, pp. 1142–1157, Mar 2011.
- [94] E. Bjornson, M. Bengtsson, and B. Ottersten, "Optimal multiuser transmit beamforming: A difficult problem with a simple solution structure [lecture notes]," *IEEE Signal Processing Magazine*, vol. 31, no. 4, pp. 142–148, Jul 2014.
- [95] D. V. B. (DVB), "Second generation framing structure, channel coding and modulation systems for broadcasting, interactive services, news gathering and other broadband satellite applications," 2014.
- [96] D. Spano, S. Chatzinotas, S. Andrenacci, J. Krause, and B. Ottersten, "Per-antenna power minimization in symbol-level precoding for the multibeam satellite downlink," *International Journal of Satellite Communications and Networking*, vol. 37, no. 1, pp. 15–30, May 2018.
- [97] D. Christopoulos, S. Chatzinotas, G. Zheng, J. Grotz, and B. Ottersten, "Linear and nonlinear techniques for multibeam joint processing in satellite communications," *EURASIP Journal on Wireless Communications and Networking*, vol. 2012, no. 1, May 2012.
- [98] P.-D. Arapoglou, K. Liolis, M. Bertinelli, A. Panagopoulos, P. Cottis, and R. D. Gaudenzi, "MIMO over satellite: A review," *IEEE Communications Surveys & Tutorials*, vol. 13, no. 1, pp. 27–51, 2011.
- [99] F. Pinto, F. Afghah, R. Radhakrishnan, and W. Edmonson, "Software defined radio implementation of DS-CDMA in inter-satellite communications for small satellites," in *2015 IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE)*, Dec 2015, pp. 1–6.
- [100] J. Li and Y. Li, "Modeling Ka-band satellite communication system with MPSK," in *2016 2nd IEEE International Conference on Computer and Communications (ICCC)*, Oct 2016, pp. 1785–1789.
- [101] A. Gaysin, V. Fadeev, and M. Hennhofer, "Survey of modulation and coding schemes for application in CubeSat systems," in *2017 Systems of Signal Synchronization, Generating and Processing in Telecommunications (SINKHROINFO)*, Jul 2017.
- [102] S. Kozłowski, K. Kurek, J. Skarzynski, K. Szczygielska, and M. Darmetko, "Investigation on adaptive satellite communication system performance using SDR technique," in *2018 22nd International Microwave and Radar Conference (MIKON)*, May 2018, pp. 363–366.
- [103] C. Hofmann, K.-U. Storek, R. T. Schwarz, and A. Knopp, "Spatial MIMO over satellite: A proof of concept," in *2016 IEEE International Conference on Communications (ICC)*, May 2016.
- [104] O. Gemikonakli and H. Aghvami, "Transmission of SDH signals through satellite channels using 8-ary PSK technique," in *IEE Colloquium on Advanced Modulation and Coding Techniques for Satellite Communications*, Jan 1992, pp. 9/1–9/4.
- [105] H. Zhang and H. Yu, "Multi-rate QC-LDPC encoder," in *2009 IEEE Circuits and Systems International Conference on Testing and Diagnosis*, 2009, pp. 1–4.
- [106] I. Recommendation, "Propagation data required for the design of earth-space land mobile telecommunication systems," *International Telecommunication Union*, pp. 681–686, 2009.
- [107] R. Prieto-Cerdeira, F. Perez-Fontan, P. Burzigotti, A. Bolea-Alamañac, and I. Sanchez-Lago, "Versatile two-state land mobile satellite channel model with first application to DVB-SH analysis," *International Journal of Satellite Communications and Networking*, vol. 28, no. 5-6, pp. 291–315, Jun 2010.
- [108] I.-R. Recommendation, "Guidelines for evaluation of radio transmission technologies for IMT-2000," *Rec. ITU-R M. 1225*, 1997.
- [109] Y. Sun, L. Yang, X. Yuan, Y. Zhang, and M. Qu, "Performance analysis of M-QAM OFDM for satellite laser communication systems," in *2020 International Wireless Communications and Mobile Computing (IWCMC)*, Jun 2020.
- [110] V. Dascal, P. Dolea, O. Cristea, and T. Palade, "Low-cost SDR-based ground receiving station for LEO satellite operations," in *2013 11th International Conference on Telecommunications in Modern Satellite, Cable and Broadcasting Services (TELSIKS)*, vol. 02, Oct 2013, pp. 627–630.
- [111] M. Bosco, P. Tortora, and D. Cinarelli, "Alma Mater ground station transceiver: A software defined radio for satellite communications," in *2014 IEEE Metrology for Aerospace (MetroAeroSpace)*, 2014, pp. 549–554.
- [112] A. Munir, N. Febrian, A. D. Setiawan, and Chairunnisa, "Signal analysis of GMSK modulation-based CubeSat automatic identification system receiver," in *2016 3rd International Conference on Information Technology, Computer, and Electrical Engineering (ICITACEE)*, 2016.
- [113] J. Skarzynski, M. Darmetko, S. Kozłowski, and K. Kurek, "SDR implementation of the receiver of adaptive communication system," *Radio Science*, vol. 51, no. 4, pp. 344–351, Apr 2016.
- [114] A. Kleinschrodt, A. Freimann, S. Christall, M. Lankl, and K. Schilling, "Advances in modulation and communication protocols for small satellite ground stations," in *Proceedings of the 68th International Astronautical Congress*, 2017, pp. 2–s2.
- [115] J. Su, Z. Lu, X. Yu, and C. Hu, "A novel low complexity soft-decision demapper for QPSK 8PSK demodulation of DVB-S2 systems," in *2011 IEEE International Conference of Electron Devices and Solid-State Circuits*. IEEE, 2011, pp. 1–2.
- [116] V. Kuznetsov, A. Malyshev, and A. Solodkov, "A method of quadrature digital modulation APSK-PAM," in *2016 International Conference on*

- Control, Instrumentation, Communication and Computational Technologies (ICCICCT)*, Dec 2016, pp. 143–146.
- [117] M. B. Mwakyanjala, É. J. Oliveira, and J. Beek, “Validation of a software-defined baseband system for satellite telemetry and telecommand,” *International Journal of Satellite Communications and Networking*, vol. 39, no. 6, pp. 605–625, May 2021.
- [118] C. C. for Space Data Systems (CCSDS), “Tm synchronization and channel coding (131.0-b-3 blue book),” 2017.
- [119] T. Synchronization, “Channel coding—summary of concept and rationale,” *Report Concerning Space Data System Standard. Informational Report CCSDS*, 2012.
- [120] M. Series, “IMT vision—framework and overall objectives of the future development of IMT for 2020 and beyond,” *Recommendation ITU*, vol. 2083, p. 21, 2015.
- [121] 3GPP, “Study on new radio (NR) to support non-terrestrial networks,” 2019.
- [122] —, “Solutions for nr to support non-terrestrial networks (NTN),” 2019.
- [123] K. Flynn, “5G in release 17—strong radio evolution (2019).”
- [124] 3GPP, “Technical specification group services and system aspects,” 2018.
- [125] W. K. Abdulwahab and A. A. Kadhim, “Comparative study of channel coding schemes for 5G,” in *2018 International Conference on Advanced Science and Engineering (ICOASE)*, 2018, pp. 239–243.
- [126] S. C. Burleigh, T. D. Cola, S. Morosi, S. Jayousi, E. Cianca, and C. Fuchs, “From connectivity to advanced internet services: A comprehensive review of small satellites communications and networks,” *Wireless Communications and Mobile Computing*, vol. 2019, pp. 1–17, May 2019.
- [127] X. Da, Y. Wang, and T. Xie, “Performance of LDPC codes for satellite communication in Ka band,” in *2009 5th International Conference on Wireless Communications, Networking and Mobile Computing*. IEEE, 2009, pp. 1–4.
- [128] C. Kun, S. Qi, L. Shengkai, and P. Chengzhi, “Implementation of encoder and decoder for LDPC codes based on FPGA,” *Journal of Systems Engineering and Electronics*, vol. 30, no. 4, p. 642, 2019.
- [129] R. Jose and A. Pe, “Analysis of hard decision and soft decision decoding algorithms of LDPC codes in AWGN,” in *2015 IEEE International Advance Computing Conference (IACC)*, 2015, pp. 430–435.
- [130] T. Richardson and R. Urbanke, “Efficient encoding of low-density parity-check codes,” *IEEE Transactions on Information Theory*, vol. 47, no. 2, pp. 638–656, 2001.
- [131] E. Sharon, S. Litsyn, and J. Goldberger, “An efficient message-passing schedule for LDPC decoding,” in *2004 23rd IEEE Convention of Electrical and Electronics Engineers in Israel*, 2004, pp. 223–226.
- [132] H. B. Thameur, B. Le Gal, N. Khouja, F. Tlili, and C. Jegou, “A survey on decoding schedules of LDPC convolutional codes and associated hardware architectures,” in *2017 IEEE Symposium on Computers and Communications (ISCC)*, 2017, pp. 898–905.
- [133] J. V. Wouterghem, A. Alloum, J. J. Boutros, and M. Moeneclaey, “On short-length error-correcting codes for 5G-NR,” *Ad Hoc Networks*, vol. 79, pp. 53–62, Oct 2018.
- [134] M. C. Coşkun, G. Durisi, T. Jerkovits, G. Liva, W. Ryan, B. Stein, and F. Steiner, “Efficient error-correcting codes in the short blocklength regime,” *Physical Communication*, vol. 34, pp. 66–79, Jun 2019.
- [135] G. Liva, L. Gaudio, T. Ninacs, and T. Jerkovits, “Code design for short blocks: A survey,” *arXiv preprint arXiv:1610.00873*, 2016.
- [136] T. Richardson and S. Kudekar, “Design of low-density parity check codes for 5G new radio,” *IEEE Communications Magazine*, vol. 56, no. 3, pp. 28–34, Mar 2018.
- [137] M. Shirvanimoghaddam, M. S. Mohammadi, R. Abbas, A. Minja, C. Yue, B. Matuz, G. Han, Z. Lin, W. Liu, Y. Li, S. Johnson, and B. Vucetic, “Short block-length codes for ultra-reliable low latency communications,” *IEEE Communications Magazine*, vol. 57, no. 2, pp. 130–137, Feb 2019.
- [138] M. Sybis, K. Wesolowski, K. Jayasinghe, V. Venkatasubramanian, and V. Vukadinovic, “Channel coding for ultra-reliable low-latency communication in 5G systems,” in *2016 IEEE 84th vehicular technology conference*, 2016, pp. 1–5.
- [139] A. Zaidi, F. Athley, J. Medbo, U. Gustavsson, G. Durisi, and X. Chen, *5G Physical Layer: principles, models and technology components*. Academic Press, 2018.
- [140] I. E. Bocharova, B. D. Kudryashov, V. Skachek, and Y. Yakimenko, “Distance properties of short LDPC codes and their impact on the BP, ML and near-ML decoding performance,” in *International Castle Meeting on Coding Theory and Applications*, 2017, pp. 48–61.
- [141] L. R. Medova, P. S. Rybin, and I. V. Filatov, “Short length LDPC code-candidate for satellite control channel,” in *2018 Engineering and Telecommunication (EnT-MIPT)*, 2018, pp. 163–166.
- [142] S. Mhaske, H. Kee, T. Ly, A. Aziz, and P. Spasojevic, “High-throughput FPGA-based QC-LDPC decoder architecture,” in *2015 IEEE 82nd Vehicular Technology Conference*, 2015, pp. 1–5.
- [143] B. Tahir, S. Schwarz, and M. Rupp, “BER comparison between convolutional, turbo, LDPC, and polar codes,” in *2017 24th international conference on telecommunications (ICT)*, 2017, pp. 1–7.
- [144] S. Lin and D. J. Costello, *Error Control Coding*. Prentice Hall, New York, 2001, vol. 2, no. 4.
- [145] H. Xu, D. Feng, C. Sun, and B. Bai, “Algebraic-based nonbinary LDPC codes with flexible field orders and code rates,” *China Communications*, vol. 14, no. 4, pp. 111–119, 2017.
- [146] Q. Diao, J. Li, S. Lin, and I. F. Blake, “New classes of partial geometries and their associated LDPC codes,” *IEEE Transactions on Information Theory*, vol. 62, no. 6, pp. 2947–2965, 2015.
- [147] B. Karimi and A. H. Banihashemi, “Construction of QC LDPC codes with low error floor by efficient systematic search and elimination of trapping sets,” *IEEE Transactions on Communications*, vol. 68, no. 2, pp. 697–712, Feb 2020.
- [148] H. Zhu, L. Pu, H. Xu, and B. Zhang, “Construction of quasi-cyclic LDPC codes based on fundamental theorem of arithmetic,” *Wireless Communications and Mobile Computing*, vol. 2018, pp. 1–9, 2018.
- [149] L. F. Abusedra, A. M. Daeri, and A. R. Zerek, “Implementation and performance study of the LDPC coding in the DVB-S2 link system using MATLAB,” in *2016 17th International Conference on Sciences and Techniques of Automatic Control and Computer Engineering (STA)*, 2016, pp. 669–674.
- [150] Q. Huang, Q. Diao, S. Lin, and K. Abdel-Ghaffar, “Cyclic and quasi-cyclic LDPC codes: new developments,” in *2011 Information Theory and Applications Workshop*, 2011, pp. 1–10.
- [151] Y. Liu and M. Zhang, “Construction of quasi-cyclic LDPC codes based on Euclidean geometries,” in *2013 8th International Conference on Communications and Networking in China (CHINACOM)*, 2013, pp. 922–924.
- [152] J. P. S. Mathur and A. Pandey, “FER performance analysis and optimization of diagonal structure based QC-LDPC codes with girth 12 using LU decomposition,” *Journal of Electrical Engineering & Technology*, vol. 15, no. 3, pp. 1405–1412, Mar 2020.
- [153] Q. Meng, J. X. Zhao, W. Xu, and L. Li, “A construction of the high-rate regular quasi-cyclic LDPC codes,” *EURASIP Journal on Wireless Communications and Networking*, vol. 2019, no. 1, Jan 2019.
- [154] H. Xu, H. Li, B. Bai, M. Zhu, and B. Zhang, “Tanner (j, l) quasi-cyclic LDPC codes: Girth analysis and derived codes,” *IEEE Access*, vol. 7, pp. 944–957, 2018.
- [155] Y. Liu and M. Zhang, “Improved quasi-cyclic LDPC codes based on Euclidean geometry,” in *2015 International Conference on Information and Communications Technologies (ICT 2015)*, 2015, pp. 1–5.
- [156] H. Xu, D. Feng, R. Luo, and B. Bai, “Construction of quasi-cyclic LDPC codes via masking with successive cycle elimination,” *IEEE Communications Letters*, vol. 20, no. 12, pp. 2370–2373, Dec 2016.
- [157] M. N. Danish, S. A. Pasha, and A. J. Hashmi, “Quasi-cyclic LDPC codes for short block-lengths,” in *2021 IEEE Aerospace Conference*, Mar 2021.
- [158] D. F. Macedo, D. Guedes, L. F. Vieira, M. A. Vieira, and M. Nogueira, “Programmable networks—from software-defined radio to software-defined networking,” *IEEE Communications Surveys & Tutorials*, vol. 17, no. 2, pp. 1102–1125, 2015.
- [159] M. R. Maheshwarappa, M. D. Bowyer, and C. P. Bridges, “Improvements in CPU & FPGA performance for small satellite SDR applications,” *IEEE Transactions on Aerospace and Electronic Systems*, vol. 53, no. 1, pp. 310–322, 2017.
- [160] —, “A reconfigurable SDR architecture for parallel satellite reception,” *IEEE Aerospace and Electronic Systems Magazine*, vol. 33, no. 11, pp. 40–53, Nov 2018.
- [161] S. Hirve and S. Gumudavally, “Wireless transmission of JPEG file using GNU radio and USRP,” *Internal unpublished project report, Department of Electrical and Computer Engineering, Cleveland State University, Cleveland, OH*, 2008.
- [162] Z. Tong, M. S. Arifanto, and C. F. Liao, “Wireless transmission using universal software radio peripheral,” in *2009 International Conference on Space Science and Communication*, Oct 2009, pp. 19–23.
- [163] S. Shearman and J. Kimery, “Software defined radio prototyping platforms enable a flexible approach to design [application notes],” *IEEE Microwave Magazine*, vol. 13, no. 5, pp. 76–80, Jul 2012.

- [164] E. Del Re, L. S. Ronga, and L. Piercucci, "Trends in satellite communications," in *2002 IEEE International Conference on Communications. Conference Proceedings*, vol. 5, 2002, pp. 2983–2988.
- [165] S. J. Olivieri, J. Aarestad, L. H. Pollard, A. M. Wyglinski, C. Kief, and R. S. Erwin, "Modular FPGA-based software defined radio for CubeSats," in *2012 IEEE International Conference on Communications (ICC)*, 2012, pp. 3229–3233.
- [166] O. Ceylan, A. Caglar, H. B. Tugrel, H. O. Cakar, A. O. Kislal, K. Kula, and H. B. Yagci, "Small satellites rock a software-defined radio modem and ground station design for cube satellite communication," *IEEE Microwave Magazine*, vol. 17, no. 3, pp. 26–33, Mar 2016.
- [167] A. O. Kislal, L. Seytanoglu, O. Cakar, K. Kula, O. Ceylan, and H. B. YAGCI, "Development of an FPGA based software defined radio transmitter for cube satellites," in *7th Nano-Satellite Symposium*, 2016.
- [168] A. Done, A. Căilean, C. Leşanu, M. Dimian, and A. Graur, "Design and implementation of a satellite communication ground station," in *2017 International Symposium on Signals, Circuits and Systems (ISSCS)*, Jul 2017, pp. 1–4.
- [169] R. P. Hudhajanto, H. Wijanarko, M. Arifin, B. Sugandi, S. Aisyah, N. S. Asaad, and A. A. Syardiansyah, "Low cost nano satellite communication system using GNURadio, HackRF, and Raspberry Pi," in *2018 International Conference on Applied Engineering (ICAE)*, 2018, pp. 1–4.
- [170] S. Kozłowski, "A carrier synchronization algorithm for SDR-based communication with LEO satellites," *Radioengineering*, vol. 27, no. 1, 2018.
- [171] A. Zainudin, A. Sudarsono, and I. G. P. Astawa, "Reliability analysis of digital communication for various data types transmission using GNU radio and USRP," in *Industrial Electronic Seminar*, 2013.
- [172] I. Lucresi, A. D. Carlofelice, and P. Tognolatti, "SDR-based system for satellite ranging measurements," *IEEE Aerospace and Electronic Systems Magazine*, vol. 31, no. 1, pp. 8–13, Jan 2016.
- [173] S. Xu, X.-W. Wang, and M. Huang, "Software-defined next-generation satellite networks: Architecture, challenges, and solutions," *IEEE Access*, vol. 6, pp. 4027–4041, 2018.
- [174] R. Purnamasari, H. Wijanto, and I. Hidayat, "Design and implementation of LDPC (low density parity check) coding technique on FPGA (field programmable gate array) for DVB-S2 (digital video broadcasting-satellite)," in *2014 IEEE International Conference on Aerospace Electronics and Remote Sensing Technology*, 2014, pp. 83–88.
- [175] D. Digdarsini, D. Mishra, S. Mehta, and T. Ram, "FPGA implementation of FEC encoder with BCH & LDPC codes for DVB S2 system," in *2019 6th International Conference on Signal Processing and Integrated Networks (SPIN)*, 2019, pp. 78–81.
- [176] E. R. De Lima, A. F. Queiroz, D. C. Alves, G. S. Da Silva, C. G. Chaves, J. G. Mertes, and T. M. Marson, "A detailed DVB-S2 receiver implementation: FPGA prototyping and preliminary ASIC resource estimation," in *2014 IEEE Latin-America Conference on Communications (LATINCOM)*, 2014, pp. 1–6.
- [177] R. Palisetty, V. K. Sinha, S. Mallick, and K. C. Ray, "FPGA prototyping of energy dispersal and improved error efficiency techniques for DVB-satellite standard," in *2015 International Conference on VLSI Systems, Architecture, Technology and Applications (VLSI-SATA)*, 2015, pp. 1–5.
- [178] P. N. TH and A. Gelgor, "Means to enhance the bandwidth gain from applying multicomponent signals in DVB-S2," in *2019 IEEE International Conference on Electrical Engineering and Photonics (EExPolytech)*, 2019, pp. 173–176.
- [179] H. C. Bui and L. Franck, "Cost effective emulation of geostationary satellite channels by means of software-defined radio," in *2014 IEEE Metrology for Aerospace (MetroAeroSpace)*, 2014, pp. 538–542.
- [180] T. M. Getu, W. Ajib, and O. A. Yeste-Ojeda, "Tensor-based efficient multi-interferer RFI excision algorithms for SIMO systems," *IEEE Transactions on Communications*, vol. 65, no. 7, pp. 3037–3052, Jul 2017.
- [181] E. Zhang, J. Zambrano, A. Amrhar, R. Landry, and W. Ajib, "Design and implementation of a wideband radio using SDR for avionic applications," in *2017 Integrated Communications, Navigation and Surveillance Conference (ICNS)*, 2017, pp. 2D2–1.
- [182] T. M. Getu, W. Ajib, O. A. Yeste-Ojeda, and R. Landry, "Tensor-based efficient multi-interferer RFI excision: Results using real-world data," in *2018 International Conference on Computing, Networking and Communications (ICNC)*, Mar 2018.
- [183] J. Duncan, J. Querol, N. Maturo, J. Krivochiza, D. Spano, N. Saba, L. Marrero, S. Chatzinotas, and B. Ottersten, "Hardware precoding demonstration in multibeam UHTS communications under realistic payload characteristics," in *Advances in Communications Satellite Systems. Proceedings of the 37th International Communications Satellite Systems Conference (ICSSC-2019)*. IET, pp. 1–17.
- [184] J. Duncan, J. Krivochiza, S. Andrenacci, S. Chatzinotas, and B. Ottersten, "Hardware demonstration of precoded communications in multi-beam UHTS systems," in *36th International Communications Satellite Systems Conference (ICSSC 2018)*, 2018, pp. 1–5.
- [185] N. Maturo, J. C. M. Duncan, J. Krivochiza, J. Querol, D. Spano, S. Chatzinotas, and B. Ottersten, "Demonstrator of precoding technique for a multi-beams satellite system," in *2019 8th International Workshop on Tracking, Telemetry and Command Systems for Space Applications (TTC)*, 2019, pp. 1–8.
- [186] F. Daneshgaran and M. Laddomada, "Transceiver front-end technology for software radio implementation of wideband satellite communication systems," *Wireless Personal Communications*, vol. 24, no. 2, pp. 99–121, 2003.
- [187] H. Hurskainen, J. Raasakka, T. Ahonen, and J. Nurmi, "Multicore software-defined radio architecture for GNSS receiver signal processing," *EURASIP Journal on Embedded Systems*, vol. 2009, pp. 1–10, 2009.
- [188] S. Sun, M. Kadoch, L. Gong, and B. Rong, "Integrating network function virtualization with SDR and SDN for 4G/5G networks," *IEEE Network*, vol. 29, no. 3, pp. 54–59, May 2015.
- [189] M. N. Danish, S. A. Pasha, and A. J. Hashmi, "Prototype design of a software-defined radio based SATCOM modem," in *2021 IEEE Aerospace Conference*, Mar 2021.
- [190] E. G. W. Peters and C. R. Benson, "A Doppler correcting software defined radio receiver design for satellite communications," *IEEE Aerospace and Electronic Systems Magazine*, vol. 35, no. 2, pp. 38–48, Feb 2020.
- [191] F. Farhangian and R. Landry, "Multi-constellation software-defined receiver for doppler positioning with LEO satellites," *Sensors*, vol. 20, no. 20, p. 5866, Oct 2020.
- [192] "Manual for icao aeronautical mobile satellite (route) service, part 2-iridium, draft v4. available online : <https://vdocuments.net/manual-for-icao-aeronautical-mobile-satellite-route-service-part-working-groups.html>."
- [193] "Ever wondered what is on the orbcomm satellite downlink? available online: <http://mdkenny.customer.netspace.net.au/orbcomm.pdf>."
- [194] Y. M. Abbas and K. Asami, "Design of software-defined radio-based adaptable packet communication system for small satellites," *Aerospace*, vol. 8, no. 6, p. 159, Jun 2021.
- [195] T. Pratt and J. E. Allnutt, *Satellite Communications*. Wiley, 2019.
- [196] M. Alsabah, M. A. Naser, B. M. Mahmmod, S. H. Abdhussain, M. R. Eissa, A. Al-Baidhani, N. K. Noordin, S. M. Sait, K. A. Al-Utaibi, and F. Hashim, "6G wireless communications networks: A comprehensive survey," *IEEE Access*, vol. 9, pp. 148 191–148 243, 2021.
- [197] M. Giordani, M. Polese, M. Mezzavilla, S. Rangan, and M. Zorzi, "Toward 6G networks: Use cases and technologies," *IEEE Communications Magazine*, vol. 58, no. 3, pp. 55–61, Mar 2020.
- [198] N. Rajatheva, I. Atzeni, E. Bjornson, A. Bourdoux, S. Buzzi, J.-B. Dore, S. Erkucuk, M. Fuentes, K. Guan, Y. Hu *et al.*, "White paper on broadband connectivity in 6G," *arXiv preprint arXiv:2004.14247*, 2020.
- [199] J.-B. Doré, R. Gerzaguet, N. Cassiau, and D. Ktenas, "Waveform contenders for 5G: Description, analysis and comparison," *Physical communication*, vol. 24, pp. 46–61, 2017.
- [200] N. Michailow, I. Gaspar, S. Krone, M. Lentmaier, and G. Fettweis, "Generalized frequency division multiplexing: Analysis of an alternative multi-carrier technique for next generation cellular systems," in *2012 International symposium on wireless communication systems (ISWCS)*, 2012, pp. 171–175.
- [201] J. Abdoli, M. Jia, and J. Ma, "Filtered OFDM: A new waveform for future wireless systems," in *2015 IEEE 16th International Workshop on Signal Processing Advances in Wireless Communications (SPAWC)*, 2015, pp. 66–70.
- [202] A. Pereira, P. Bento, M. Gomes, R. Dinis, and V. Silva, "TIBWB-OFDM: A promising modulation technique for MIMO 5G transmissions," in *2018 IEEE 88th Vehicular Technology Conference*, 2018, pp. 1–5.
- [203] F. Conceição, M. Gomes, V. Silva, and R. Dinis, "Highly efficient TIBWB-OFDM waveform for broadband wireless communications," in *2020 IEEE 91st Vehicular Technology Conference*, 2020, pp. 1–5.
- [204] L. Fay, L. Michael, D. Gomez-Barquero, N. Ammar, and M. W. Caldwell, "An overview of the ATSC 3.0 physical layer specification," *IEEE Transactions on Broadcasting*, vol. 62, no. 1, pp. 159–171, Mar 2016.

- [205] M. Caus, M. Á. Vázquez, and A. Pérez-Neira, "NOMA and interference limited satellite scenarios," in *2016 50th Asilomar Conference on Signals, Systems and Computers*, 2016, pp. 497–501.
- [206] N. A. K. Beigi and M. R. Soleymani, "Interference management using cooperative NOMA in multi-beam satellite systems," in *2018 IEEE International Conference on Communications (ICC)*, 2018, pp. 1–6.
- [207] X. Yan, H. Xiao, K. An, G. Zheng, and W. Tao, "Hybrid satellite terrestrial relay networks with cooperative non-orthogonal multiple access," *IEEE Communications Letters*, vol. 22, no. 5, pp. 978–981, May 2018.
- [208] Y. Sun, Z. Ding, X. Dai, and O. A. Dobre, "On the performance of network NOMA in uplink CoMP systems: A stochastic geometry approach," *IEEE Transactions on Communications*, vol. 67, no. 7, pp. 5084–5098, Jul 2019.
- [209] M. S. Ali, E. Hossain, A. Al-Dweik, and D. I. Kim, "Downlink power allocation for CoMP-NOMA in multi-cell networks," *IEEE Transactions on Communications*, vol. 66, no. 9, pp. 3982–3998, Sep 2018.
- [210] Y. Dai, J. Liu, M. Sheng, N. Cheng, and X. Shen, "Joint optimization of BS clustering and power control for NOMA-enabled CoMP transmission in dense cellular networks," *IEEE Transactions on Vehicular Technology*, vol. 70, no. 2, pp. 1924–1937, feb 2021.
- [211] M. Hedayati and I.-M. Kim, "CoMP-NOMA in the SWIPT networks," *IEEE Transactions on Wireless Communications*, vol. 19, no. 7, pp. 4549–4562, Jul 2020.
- [212] L. Liu, S. Zhang, and R. Zhang, "CoMP in the sky: UAV placement and movement optimization for multi-user communications," *IEEE Transactions on Communications*, vol. 67, no. 8, pp. 5645–5658, Aug 2019.
- [213] J. Yao, C. Zhong, Z. Liu, and J. Xu, "3d trajectory optimization for secure UAV communication with CoMP reception," in *2019 IEEE Global Communications Conference (GLOBECOM)*, 2019, pp. 1–6.
- [214] A. Kilzi, J. Farah, C. A. Nour, and C. Douillard, "Analysis of drone placement strategies for complete interference cancellation in two-cell NOMA CoMP systems," *IEEE Access*, vol. 8, pp. 179 055–179 069, 2020.
- [215] A. Wang, L. Lei, E. Lagunas, A. I. P. Neira, S. Chatzinotas, and B. Ottersten, "On fairness optimization for NOMA-enabled multi-beam satellite systems," in *2019 IEEE 30th Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, Sep 2019.
- [216] Y. Liu, S. Zhang, Z. Ding, R. Schober, N. Al-Dhahir, E. Hossain, and X. Shen, "Guest editorial special issue on next generation multiple access—part II," *IEEE Journal on Selected Areas in Communications*, vol. 40, no. 5, pp. 1387–1391, May 2022.
- [217] J. Liu and X. Wang, "Unsources multiple access based on sparse tanner graph—efficient decoding, analysis, and optimization," *IEEE Journal on Selected Areas in Communications*, vol. 40, no. 5, pp. 1509–1521, May 2022.
- [218] S. Yue, J. Ren, J. Xin, D. Zhang, Y. Zhang, and W. Zhuang, "Efficient federated meta-learning over multi-access wireless networks," *IEEE Journal on Selected Areas in Communications*, vol. 40, no. 5, pp. 1556–1570, May 2022.
- [219] Z. Guo, Z. Chen, P. Liu, J. Luo, X. Yang, and X. Sun, "Multi-agent reinforcement learning-based distributed channel access for next generation wireless networks," *IEEE Journal on Selected Areas in Communications*, vol. 40, no. 5, pp. 1587–1599, May 2022.
- [220] R. Li, H. Yan, and D. Cabric, "Rainbow-link: Beam-alignment-free and grant-free mmW multiple access using true-time-delay array," *IEEE Journal on Selected Areas in Communications*, vol. 40, no. 5, pp. 1692–1705, May 2022.
- [221] F. Xing, S. He, V. C. Leung, and H. Yin, "Energy efficiency optimization for rate-splitting multiple access-based indoor visible light communication networks," *IEEE Journal on Selected Areas in Communications*, vol. 40, no. 5, pp. 1706–1720, May 2022.
- [222] R. M. Calvo, J. Poliak, J. Surof, A. Reeves, M. Richerzhagen, H. F. Kelemu, R. Barrios, C. Carrizo, R. Wolf, F. Rein *et al.*, "Optical technologies for very high throughput satellite communications," in *Free-Space Laser Communications XXXI*, vol. 10910. SPIE, 2019, pp. 189–204.
- [223] E. Lutz, "Towards the terabit/s satellite - interference issues in the user link," *International Journal of Satellite Communications and Networking*, vol. 34, no. 4, pp. 461–482, Jun 2015.
- [224] J. Poliak, R. M. Calvo, and F. Rein, "Demonstration of 1.72 Tbit/s optical data transmission under worst-case turbulence conditions for ground-to-geostationary satellite communications," *IEEE Communications Letters*, vol. 22, no. 9, pp. 1818–1821, Sep 2018.
- [225] X. Huang, J. A. Zhang, R. P. Liu, Y. J. Guo, and L. Hanzo, "Airplane-aided integrated networking for 6G wireless: Will it work?" *IEEE Vehicular Technology Magazine*, vol. 14, no. 3, pp. 84–91, Sep 2019.
- [226] Z. Zhang, Y. Xiao, Z. Ma, M. Xiao, Z. Ding, X. Lei, G. K. Karagiannis, and P. Fan, "6G wireless networks: Vision, requirements, architecture, and key technologies," *IEEE Vehicular Technology Magazine*, vol. 14, no. 3, pp. 28–41, Sep 2019.
- [227] R. Hadani, S. Rakib, M. Tsatsanis, A. Monk, A. J. Goldsmith, A. F. Molisch, and R. Calderbank, "Orthogonal time frequency space modulation," in *2017 IEEE Wireless Communications and Networking Conference (WCNC)*, 2017, pp. 1–6.
- [228] M. I. Qadri and M. Zia, "Adaptive modulation and coding with selective retransmission under OFDM signaling," *Wireless Personal Communications*, vol. 101, no. 4, pp. 1787–1805, Jul 2018.
- [229] J. Huang, Y. Su, W. Liu, and F. Wang, "Adaptive modulation and coding techniques for global navigation satellite system inter-satellite communication based on the channel condition," *IET Communications*, vol. 10, no. 16, pp. 2091–2095, Nov 2016.
- [230] R. Xue, H. Yu, and Q. Cheng, "Adaptive coded modulation based on continuous phase modulation for inter-satellite links of global navigation satellite systems," *IEEE Access*, vol. 6, pp. 20 652–20 662, 2018.
- [231] H. Gamage, N. Rajatheva, and M. Latva-Aho, "Channel coding for enhanced mobile broadband communication in 5G systems," in *2017 European Conference on Networks and Communications (EuCNC)*, 2017, pp. 1–6.
- [232] T. T. B. Nguyen, T. N. Tan, and H. Lee, "Efficient QC-LDPC encoder for 5G new radio," *Electronics*, vol. 8, no. 6, p. 668, Jun 2019.
- [233] X. Wu, M. Jiang, C. Zhao, L. Ma, and Y. Wei, "Low-rate PBRL-LDPC codes for URLLC in 5G," *IEEE Wireless Communications Letters*, vol. 7, no. 5, pp. 800–803, Oct 2018.
- [234] A. Sharma and M. Salim, "On the feasibility of polar code as channel code candidate for the 5g-IoT scenarios," *International Journal of Future Generation Communication and Networking*, vol. 11, no. 3, pp. 11–20, May 2018.
- [235] H. Zhang, R. Li, J. Wang, S. Dai, G. Zhang, Y. Chen, H. Luo, and J. Wang, "Parity-check polar coding for 5G and beyond," in *2018 IEEE International Conference on Communications (ICC)*, 2018, pp. 1–7.
- [236] X. Xu, S. Wu, D. Dong, J. Jiao, and Q. Zhang, "High performance short polar codes: A concatenation scheme using spinal codes as the outer code," *IEEE Access*, vol. 6, pp. 70 644–70 654, 2018.
- [237] T. Koike-Akino, C. Cao, and Y. Wang, "Turbo product codes with irregular polar coding for high-throughput parallel decoding in wireless OFDM transmission," in *2018 IEEE International Conference on Communications (ICC)*, 2018, pp. 1–7.
- [238] X. Zhang, S. Chen, and Y. Liu, "Irregular LDPC-polar concatenation scheme with designed scaling factors," *Electronics Letters*, vol. 55, no. 1, pp. 57–59, Jan 2019.
- [239] H. Bian, R. Liu, A. Kaushik, Y. Hu, and J. S. Thompson, "Design of segmented CRC-aided spinal codes for IoT applications," *IET Communications*, vol. 14, no. 20, pp. 3541–3548, Nov 2020.
- [240] Y. Polyanskiy, H. V. Poor, and S. Verdú, "Channel coding rate in the finite blocklength regime," *IEEE Transactions on Information Theory*, vol. 56, no. 5, pp. 2307–2359, May 2010.
- [241] H. Chen, R. Abbas, P. Cheng, M. Shirvanimoghaddam, W. Hardjawana, W. Bao, Y. Li, and B. Vucetic, "Ultra-reliable low latency cellular networks: Use cases, challenges and approaches," *IEEE Communications Magazine*, vol. 56, no. 12, pp. 119–125, Dec 2018.
- [242] L. Xiang, M. F. Brejza, R. G. Maunder, B. M. Al-Hashimi, and L. Hanzo, "Arbitrarily parallel turbo decoding for ultra-reliable low latency communication in 3GPP LTE," *IEEE Journal on Selected Areas in Communications*, vol. 37, no. 4, pp. 826–838, Apr 2019.
- [243] S. Weithoffer, O. Griebel, R. Klaimi, C. A. Nour, and N. Wehn, "Advanced hardware architectures for turbo code decoding beyond 100 Gb/s," in *2020 IEEE Wireless Communications and Networking Conference (WCNC)*, 2020, pp. 1–6.
- [244] J. Krivochiza, J. C. Merlano-Duncan, S. Chatzinotas, and B. Ottersten, "M-QAM modulation symbol-level precoding for power minimization: Closed-form solution," in *2019 16th International Symposium on Wireless Communication Systems (ISWCS)*, 2019, pp. 395–399.
- [245] J. Krivochiza, J. M. Duncan, S. Andrenacci, S. Chatzinotas, and B. Ottersten, "FPGA acceleration for computationally efficient symbol-level precoding in multi-user multi-antenna communication systems," *IEEE Access*, vol. 7, pp. 15 509–15 520, 2019.